

BFS Traversal

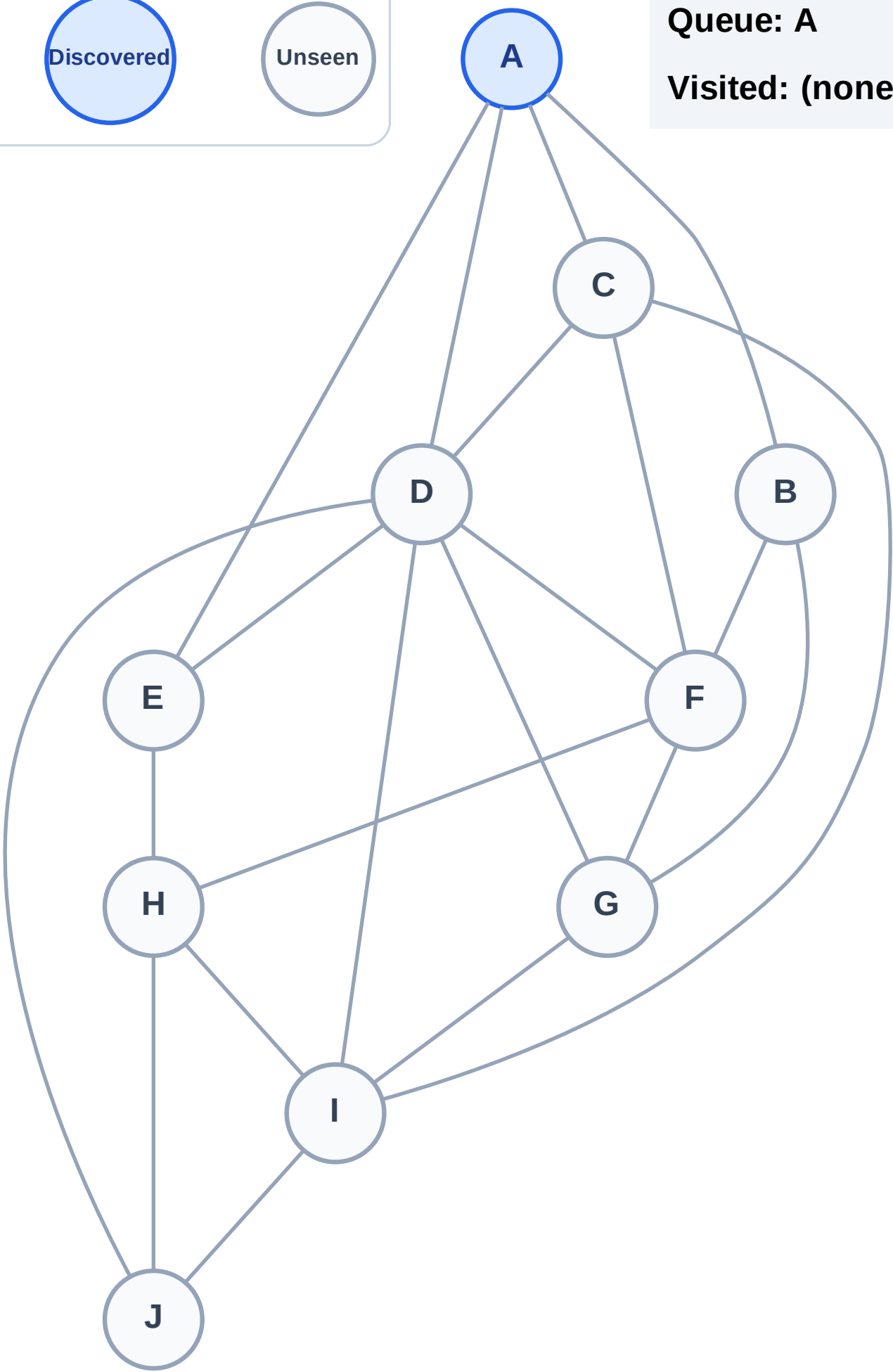
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

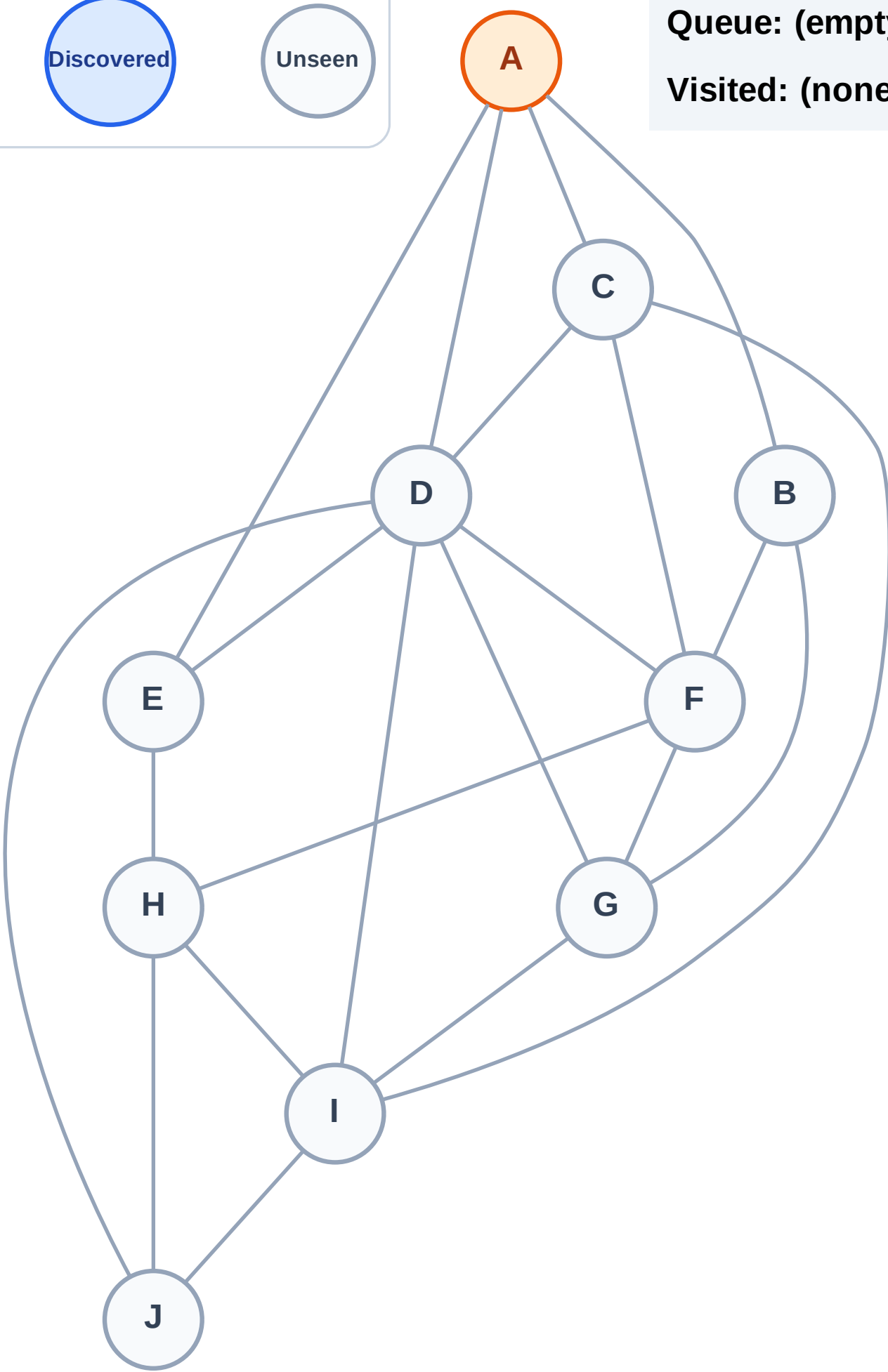
Step 2: Process node A

Legend



Queue: (empty)

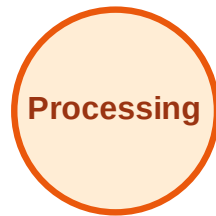
Visited: (none)



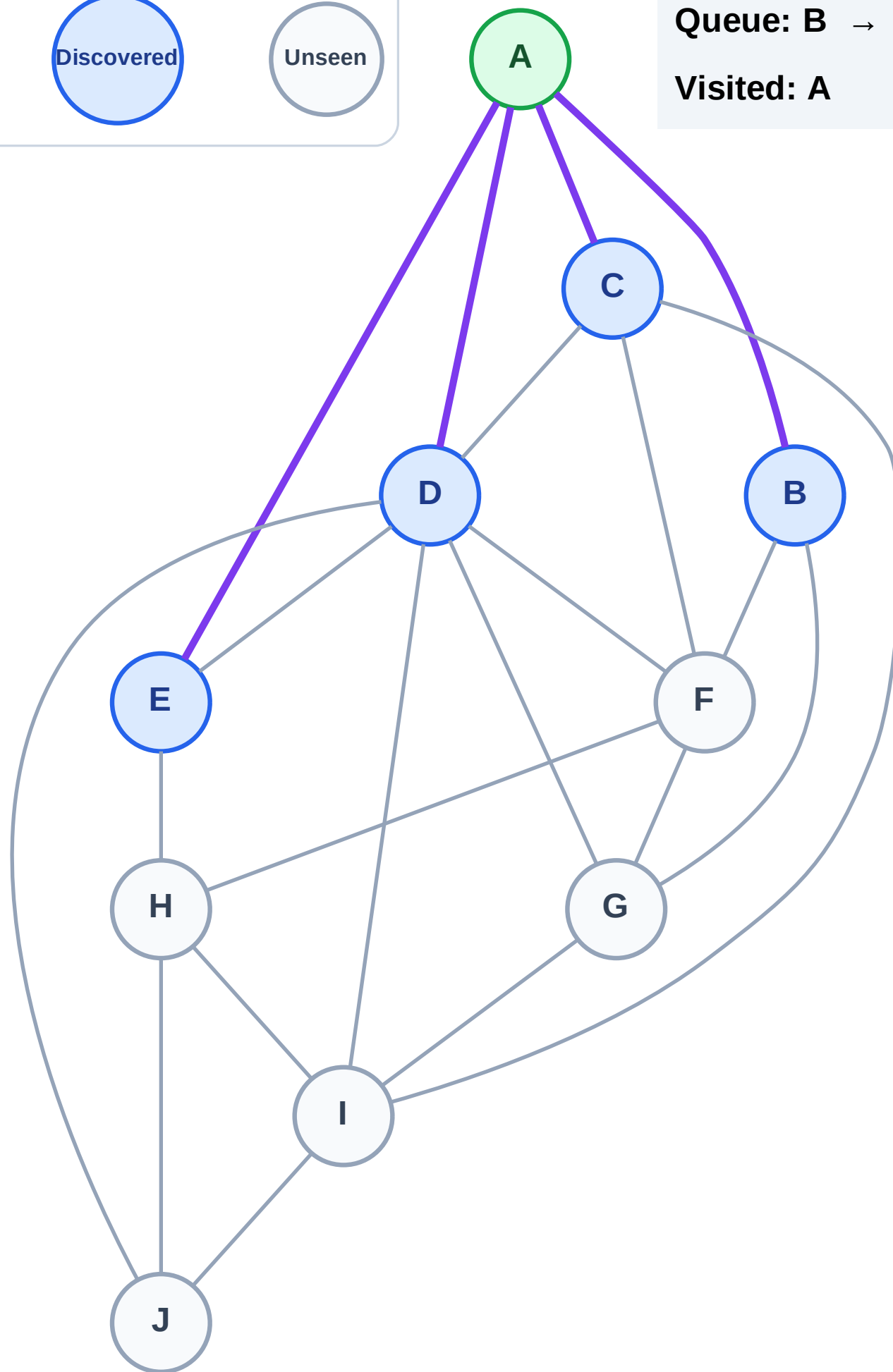
BFS Traversal

Step 3: Discover B, C, D, E from A

Legend



Queue: B → C → D → E
Visited: A



BFS Traversal

Step 4: Process node B

Legend

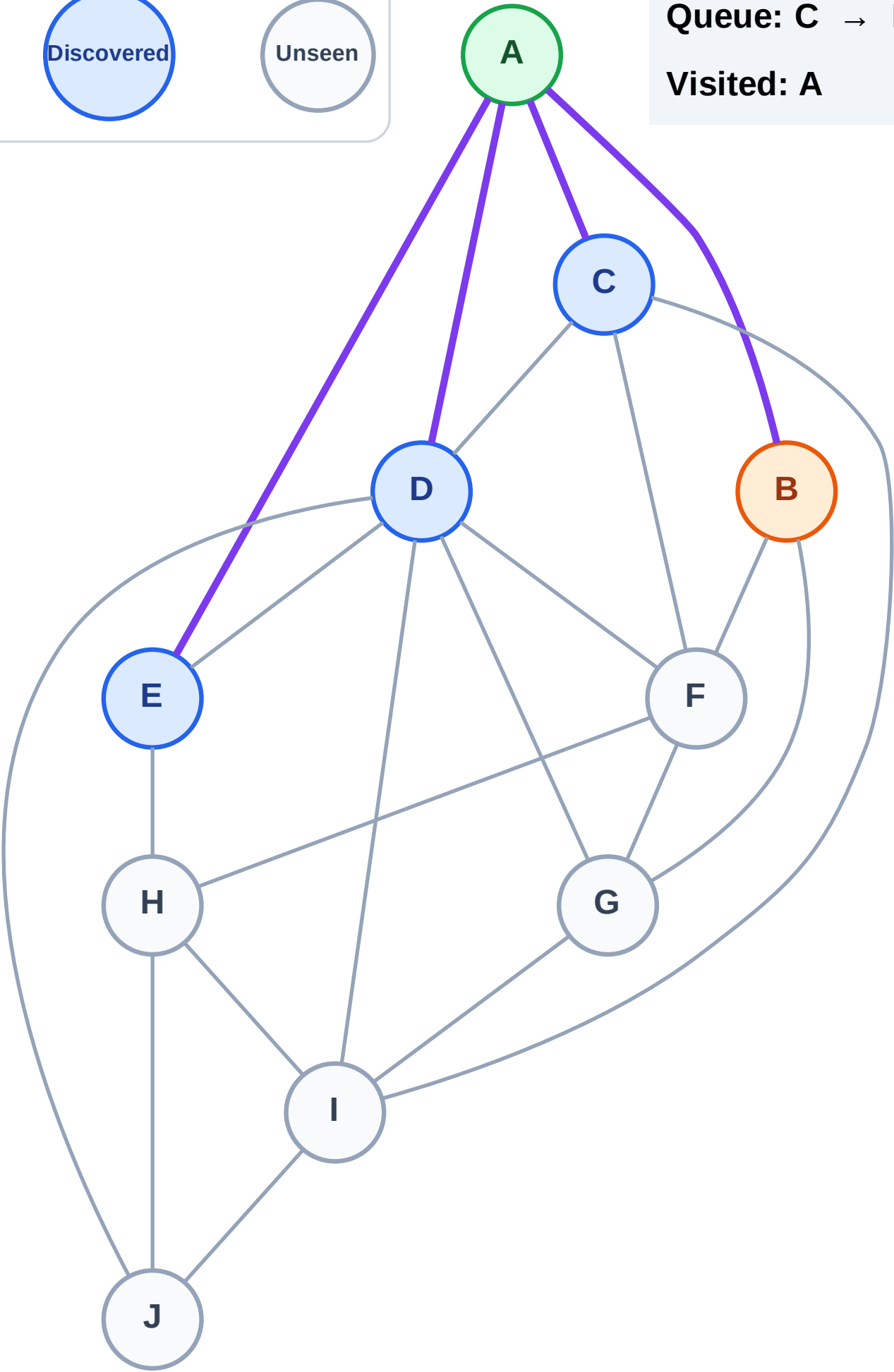
Complete

Processing

Discovered

Unseen

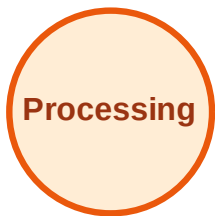
Queue: C → D → E
Visited: A



BFS Traversal

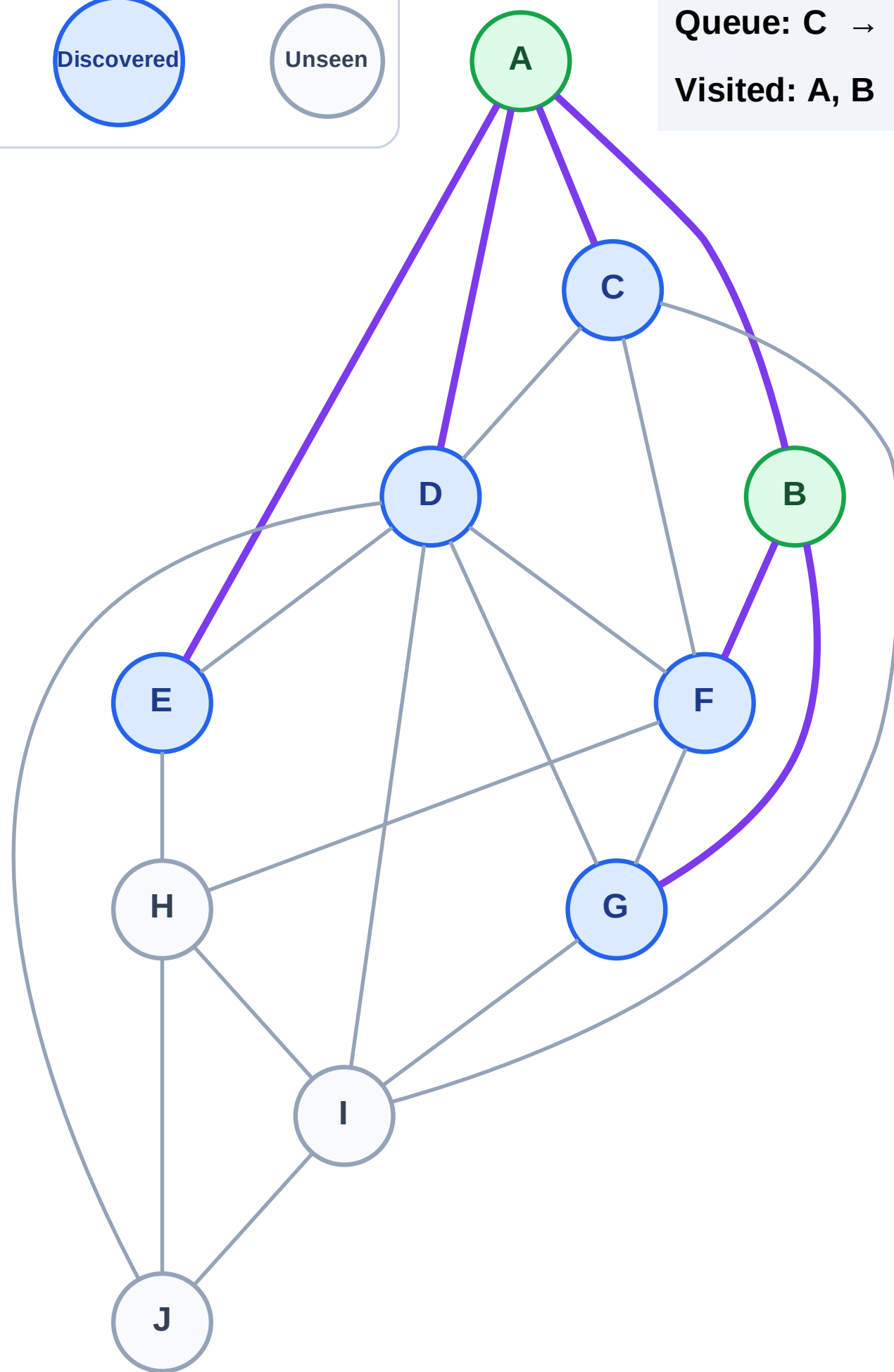
Step 5: Discover F, G from B

Legend



Queue: C → D → E → F → G

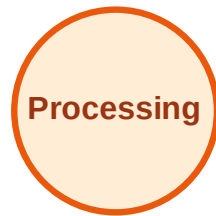
Visited: A, B



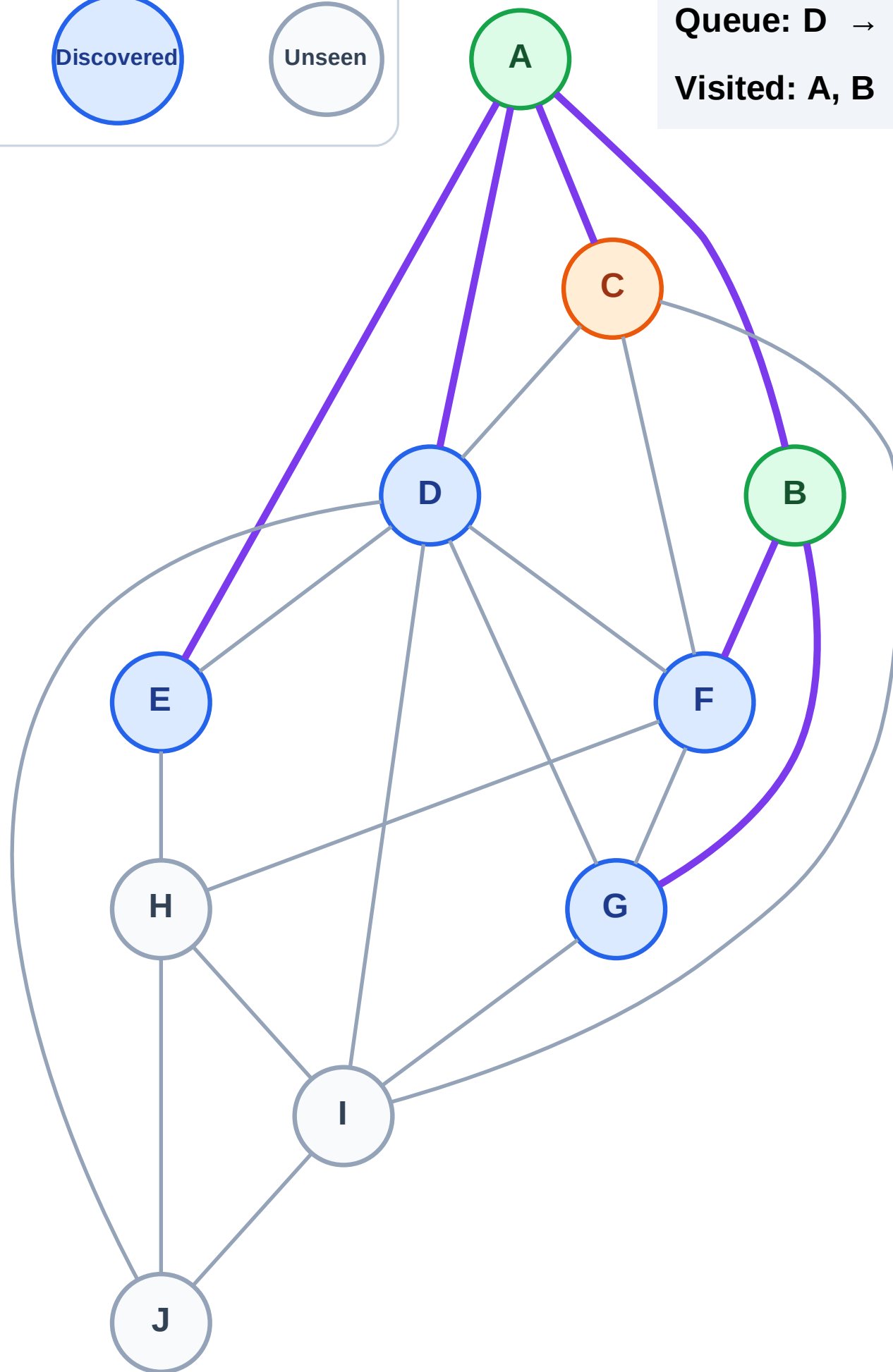
BFS Traversal

Step 6: Process node C

Legend



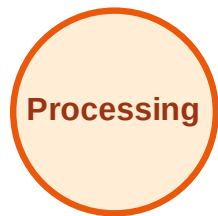
Queue: D → E → F → G
Visited: A, B



BFS Traversal

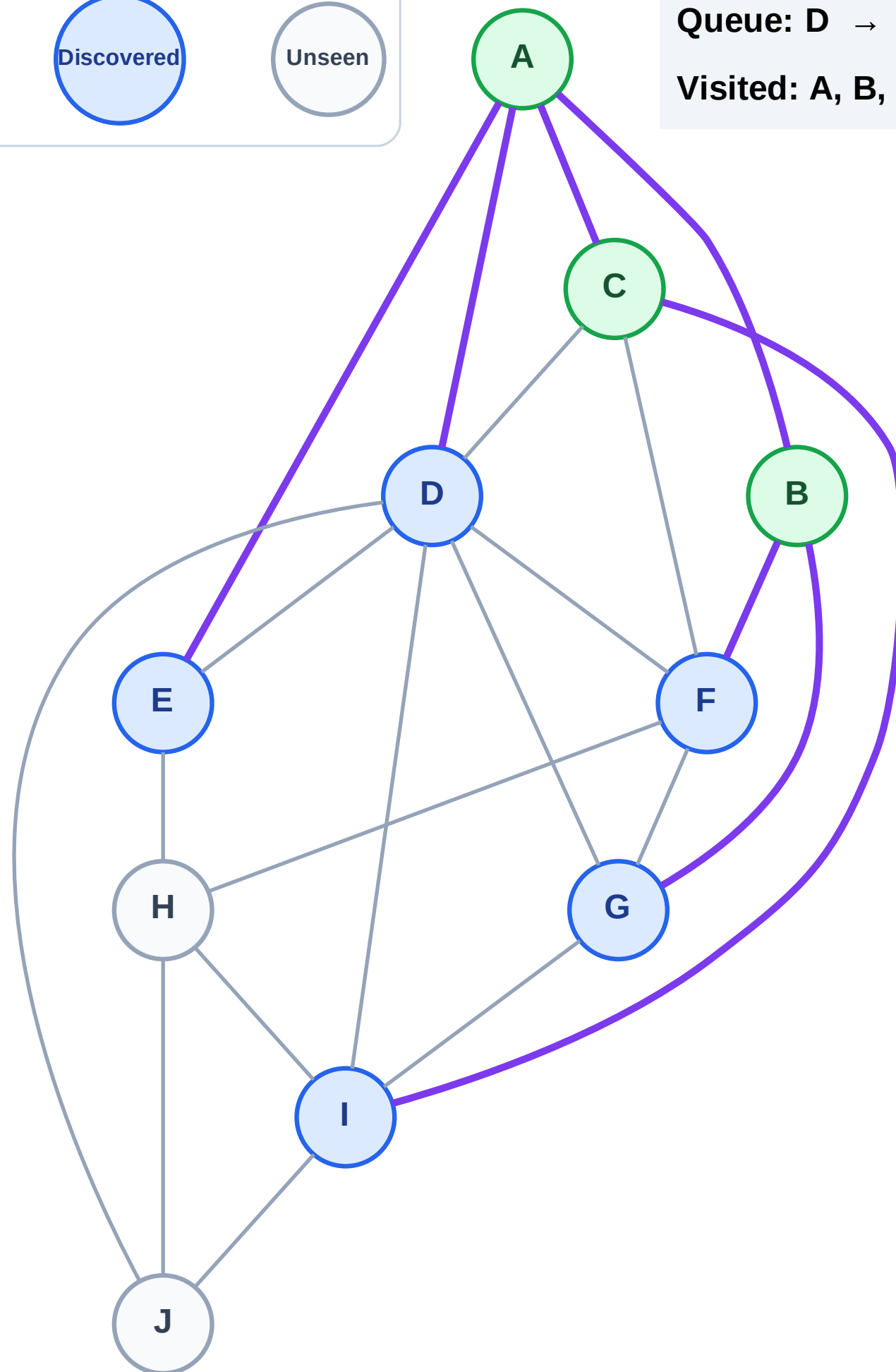
Step 7: Discover I from C

Legend



Queue: D → E → F → G → I

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

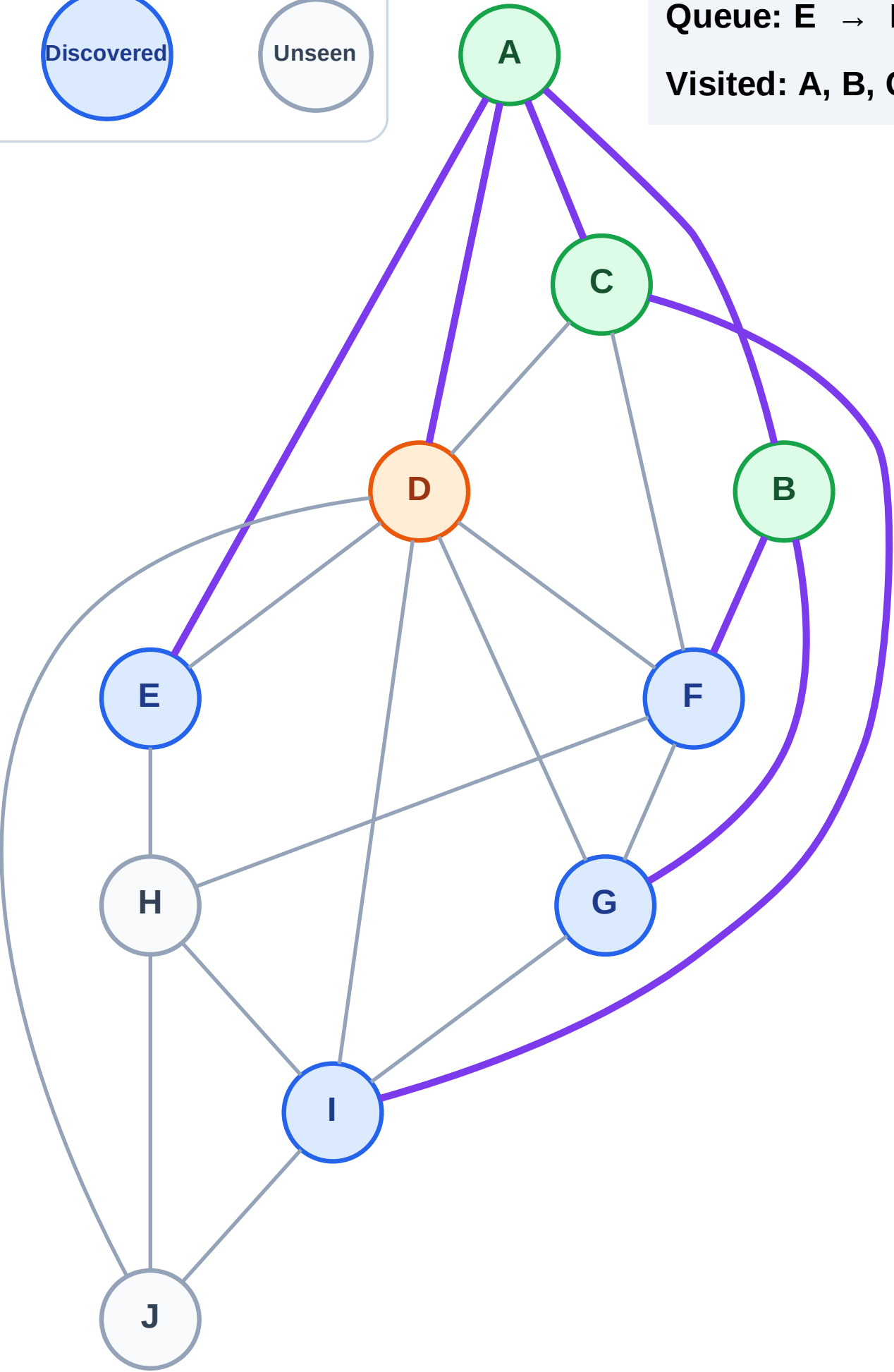
Processing

Discovered

Unseen

Queue: E → F → G → I

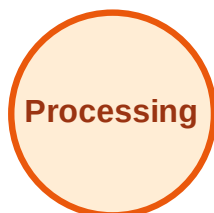
Visited: A, B, C



BFS Traversal

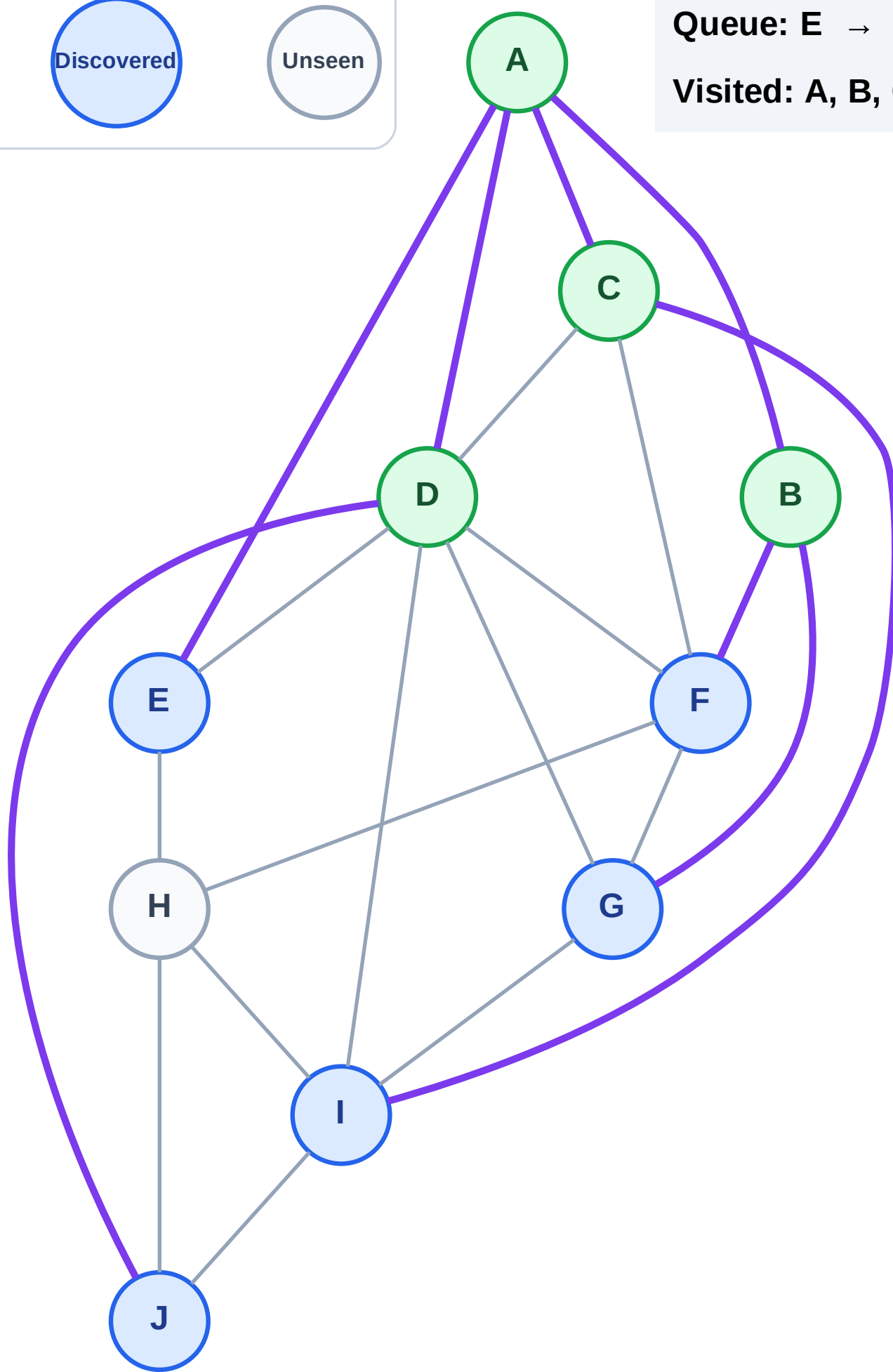
Step 9: Discover J from D

Legend



Queue: E → F → G → I → J

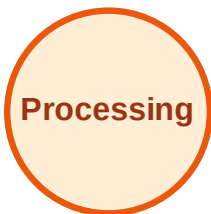
Visited: A, B, C, D



BFS Traversal

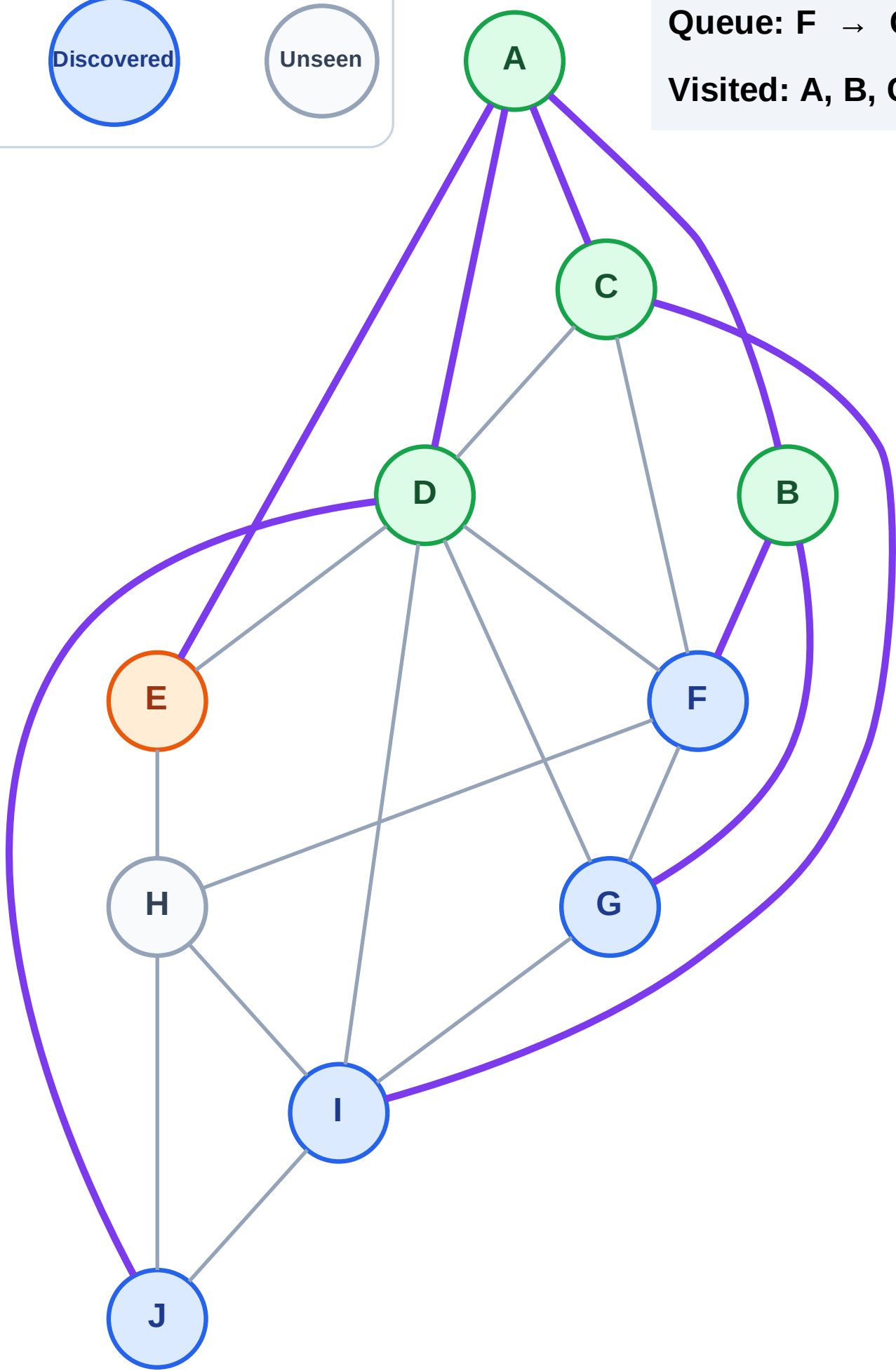
Step 10: Process node E

Legend



Queue: F → G → I → J

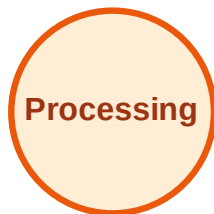
Visited: A, B, C, D



BFS Traversal

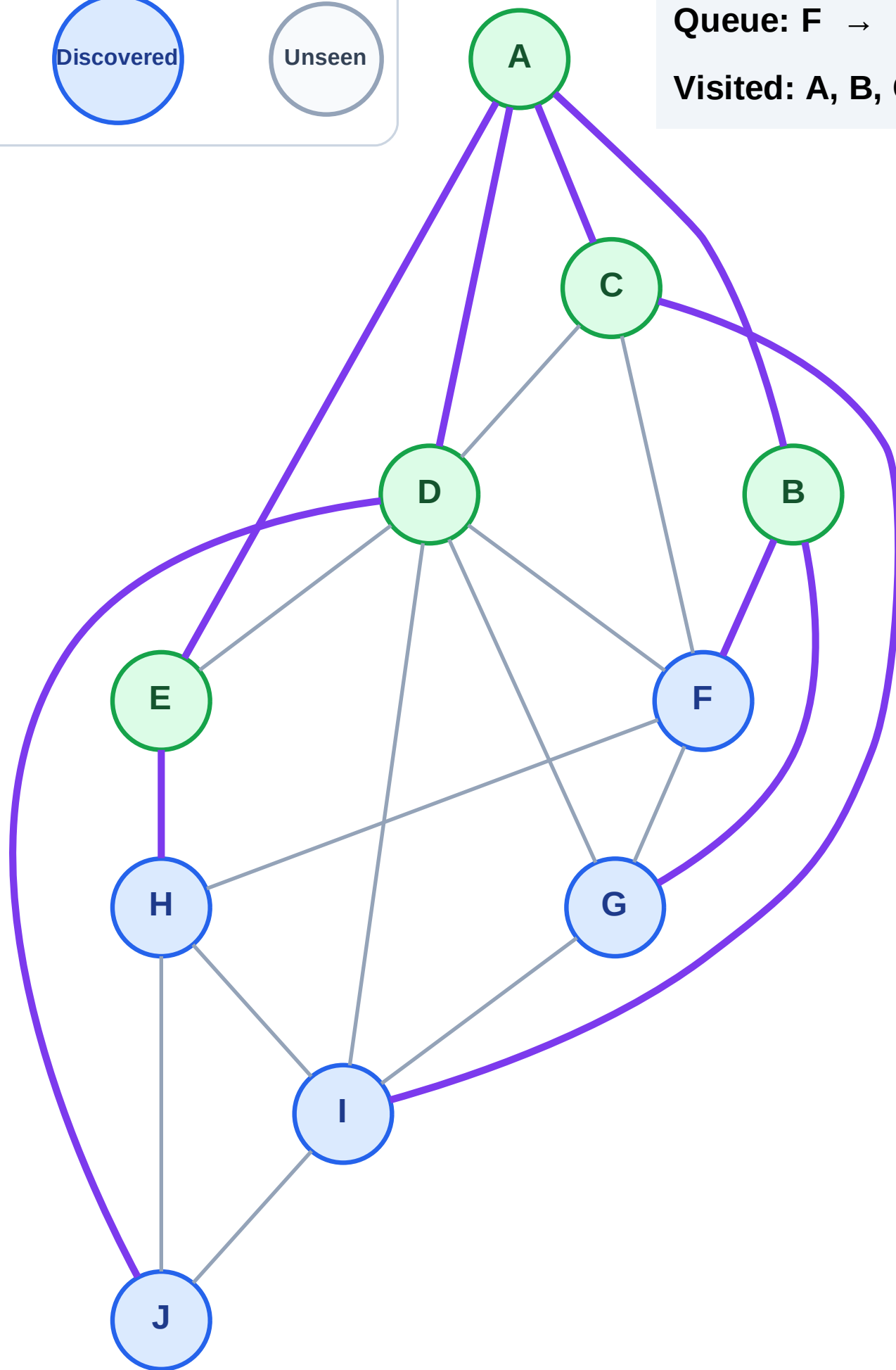
Step 11: Discover H from E

Legend



Queue: F → G → I → J → H

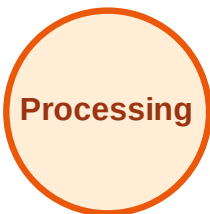
Visited: A, B, C, D, E



BFS Traversal

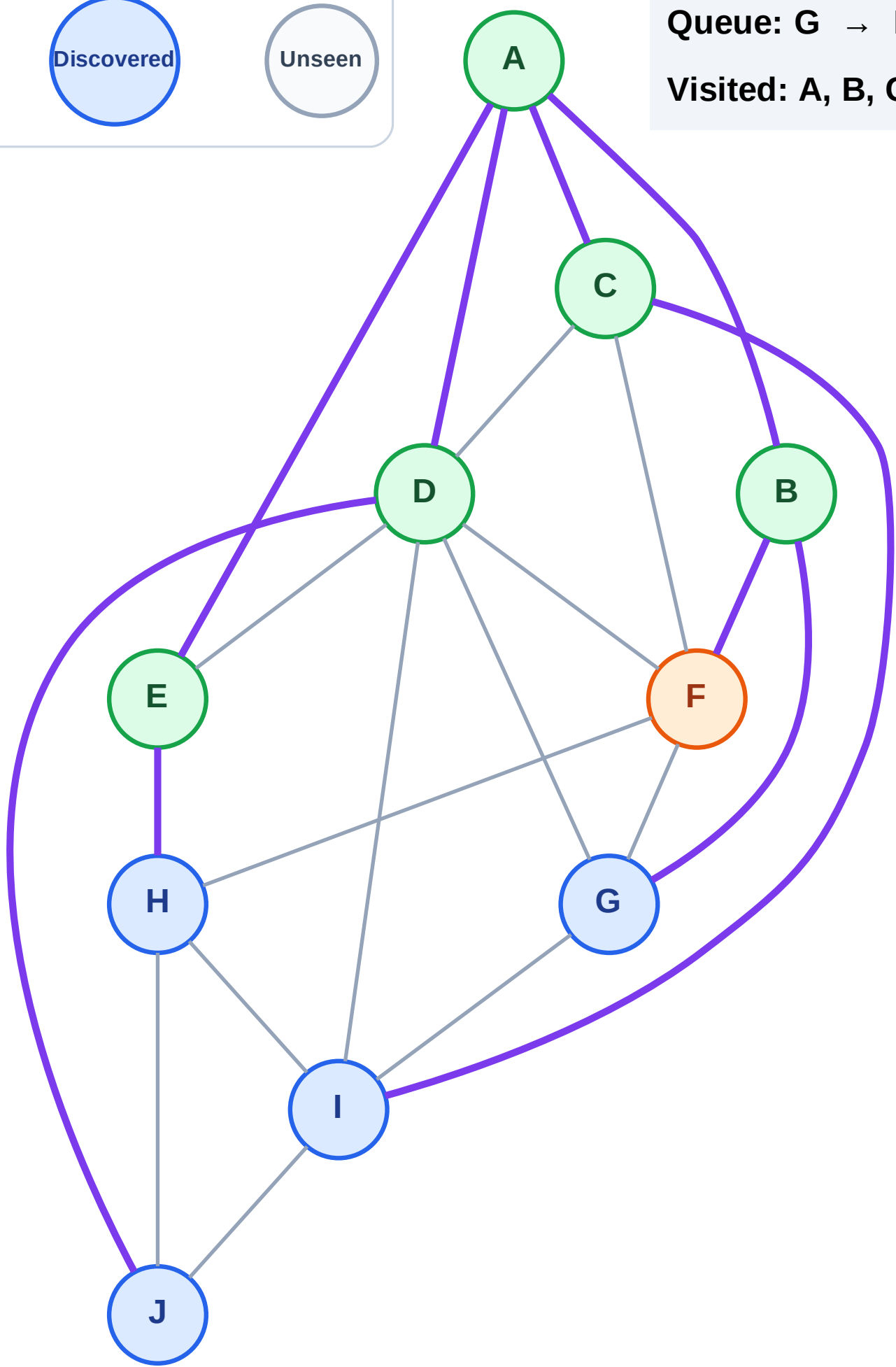
Step 12: Process node F

Legend



Queue: G → I → J → H

Visited: A, B, C, D, E



BFS Traversal

Step 13: No new neighbors from F

Legend

Complete

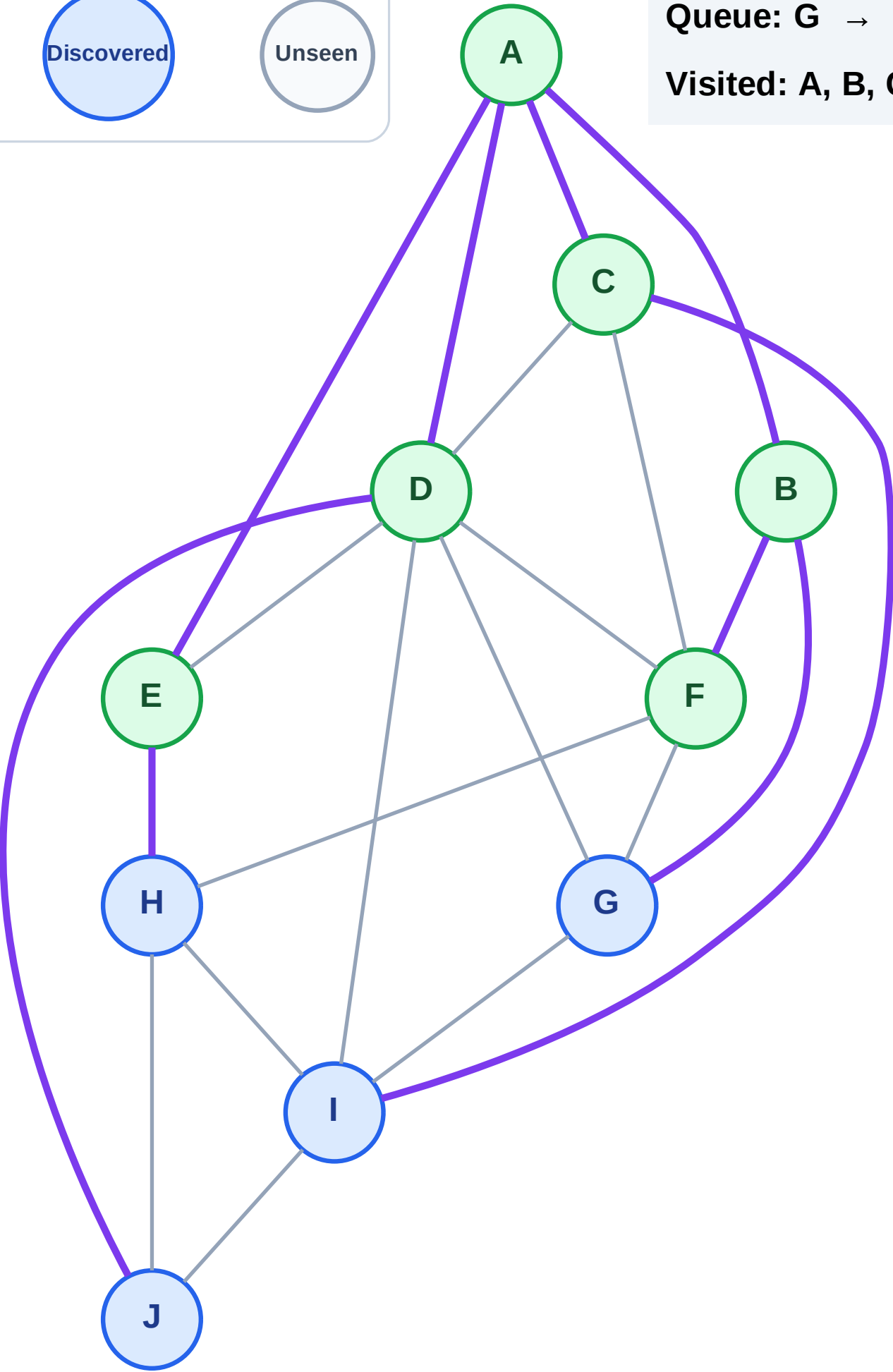
Processing

Discovered

Unseen

Queue: G → I → J → H

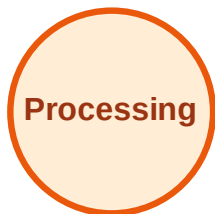
Visited: A, B, C, D, E, F



BFS Traversal

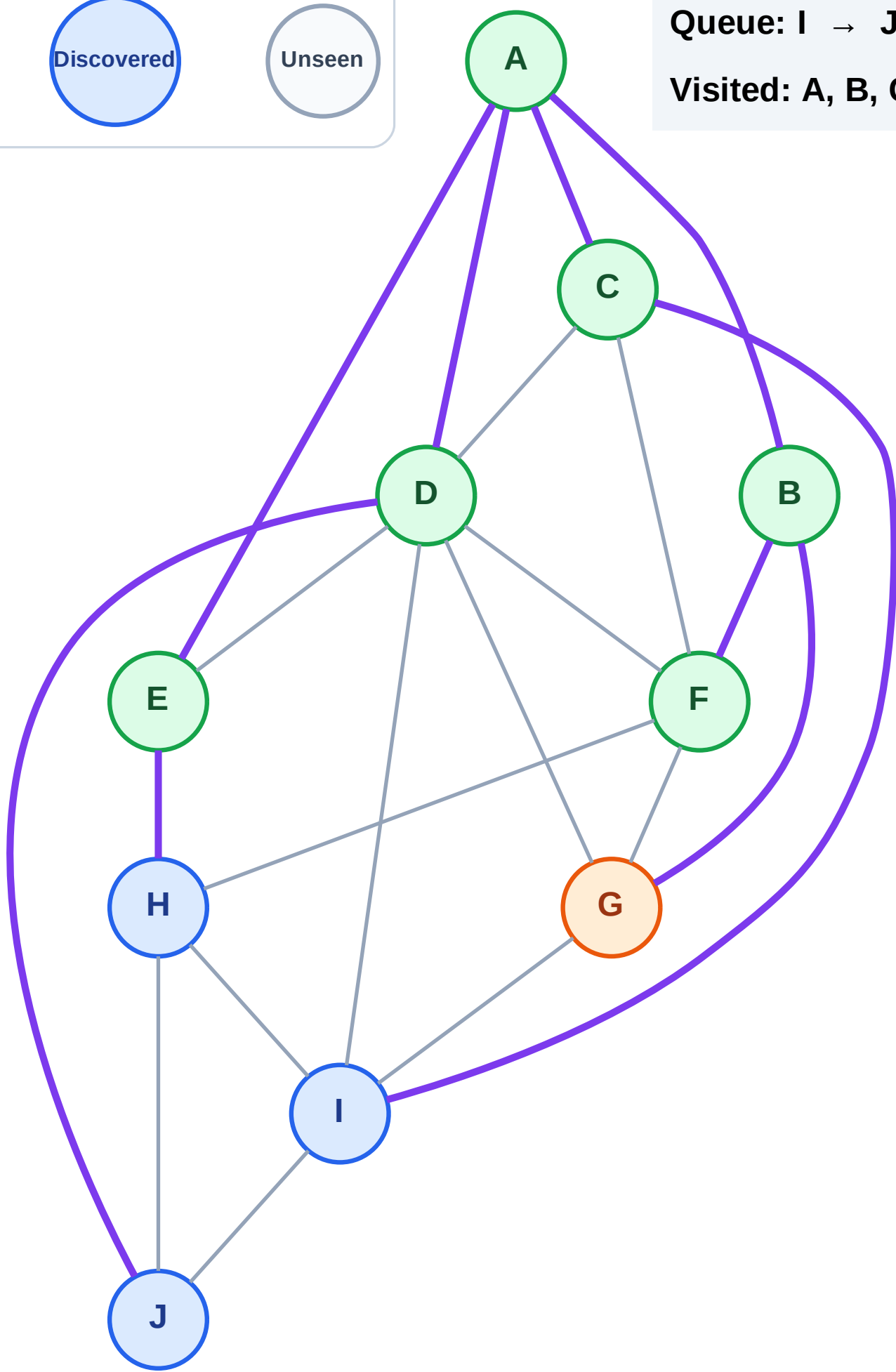
Step 14: Process node G

Legend



Queue: I → J → H

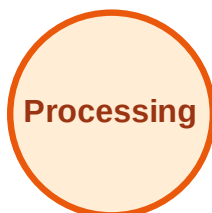
Visited: A, B, C, D, E, F



BFS Traversal

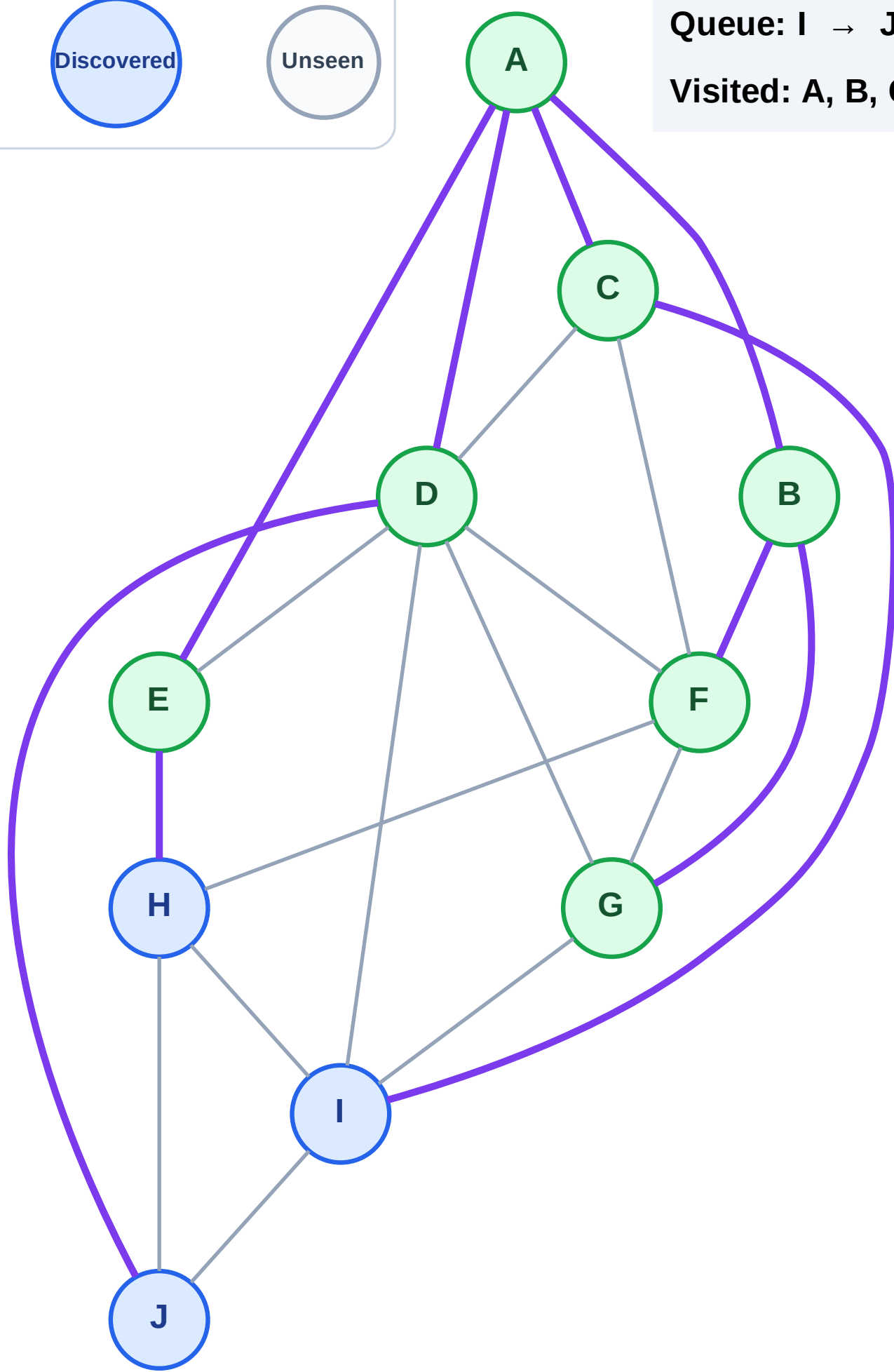
Step 15: No new neighbors from G

Legend

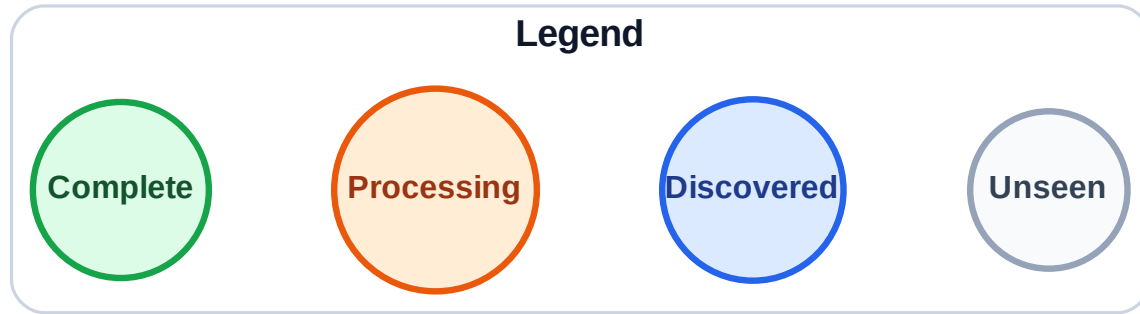


Queue: I → J → H

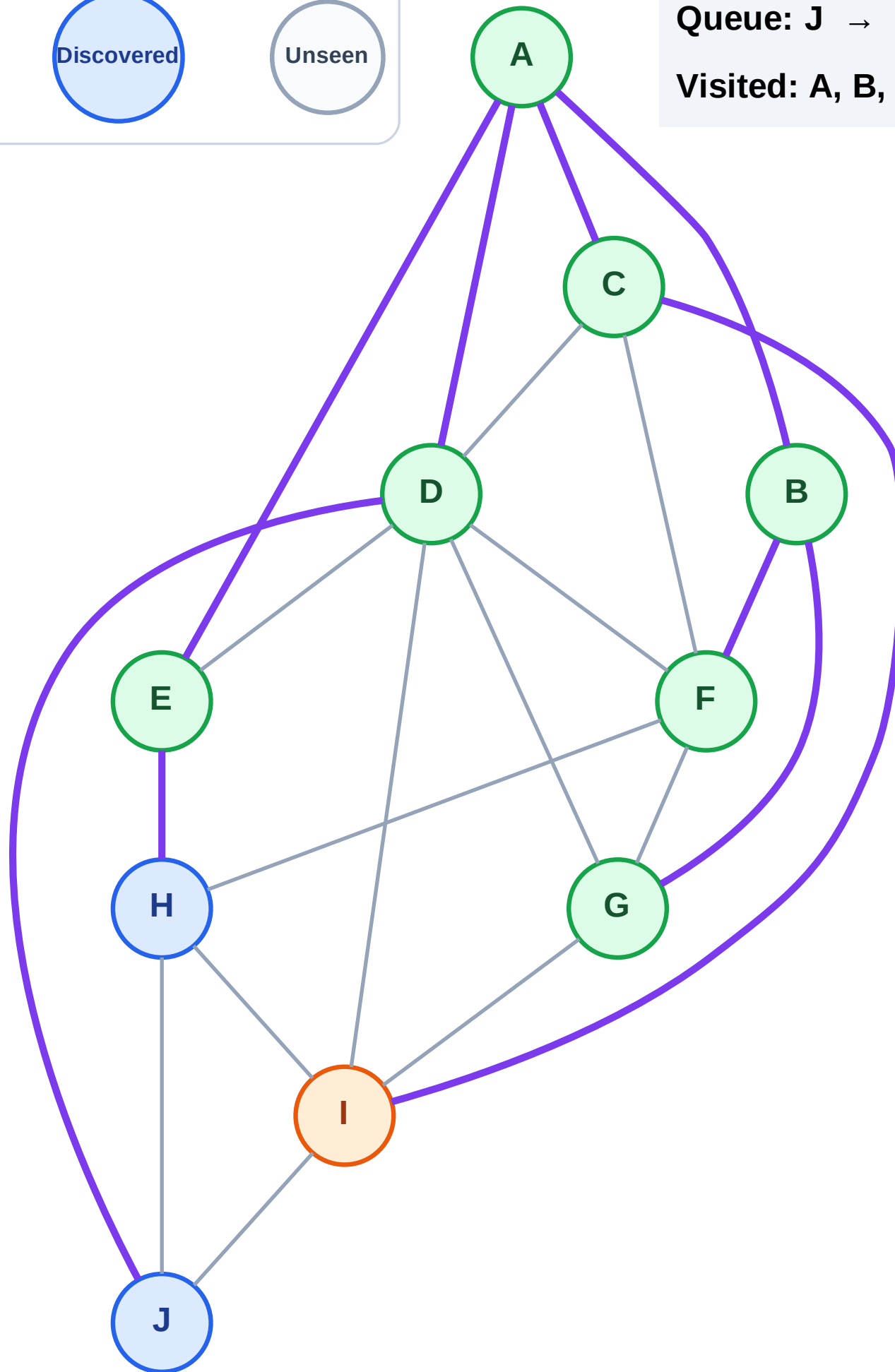
Visited: A, B, C, D, E, F, G



Step 16: Process node I



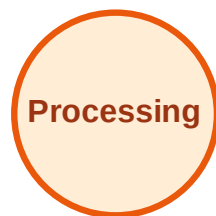
Visited: A, B, C, D, E, F, G



BFS Traversal

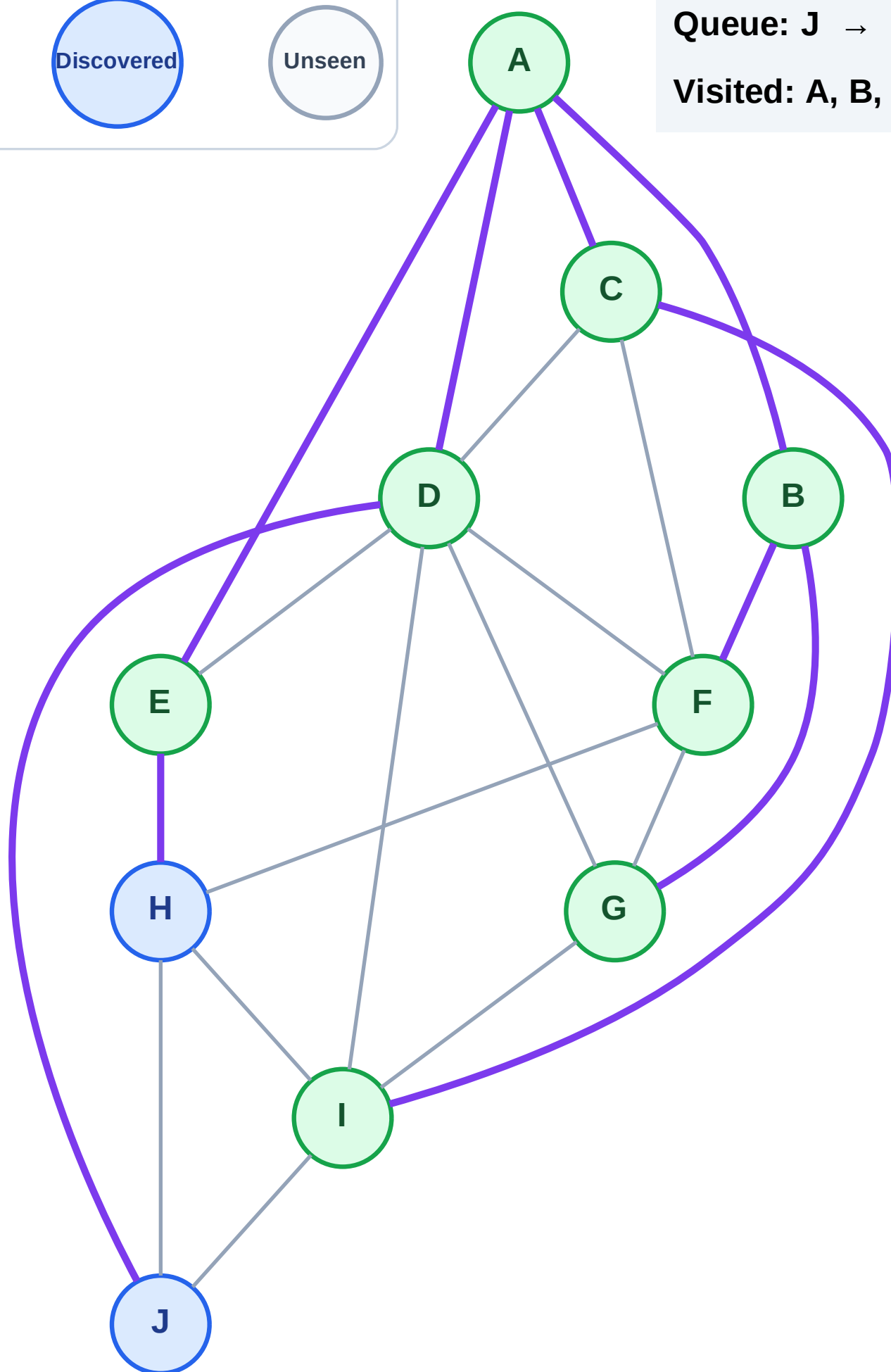
Step 17: No new neighbors from I

Legend



Queue: J → H

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node J

Legend

Complete

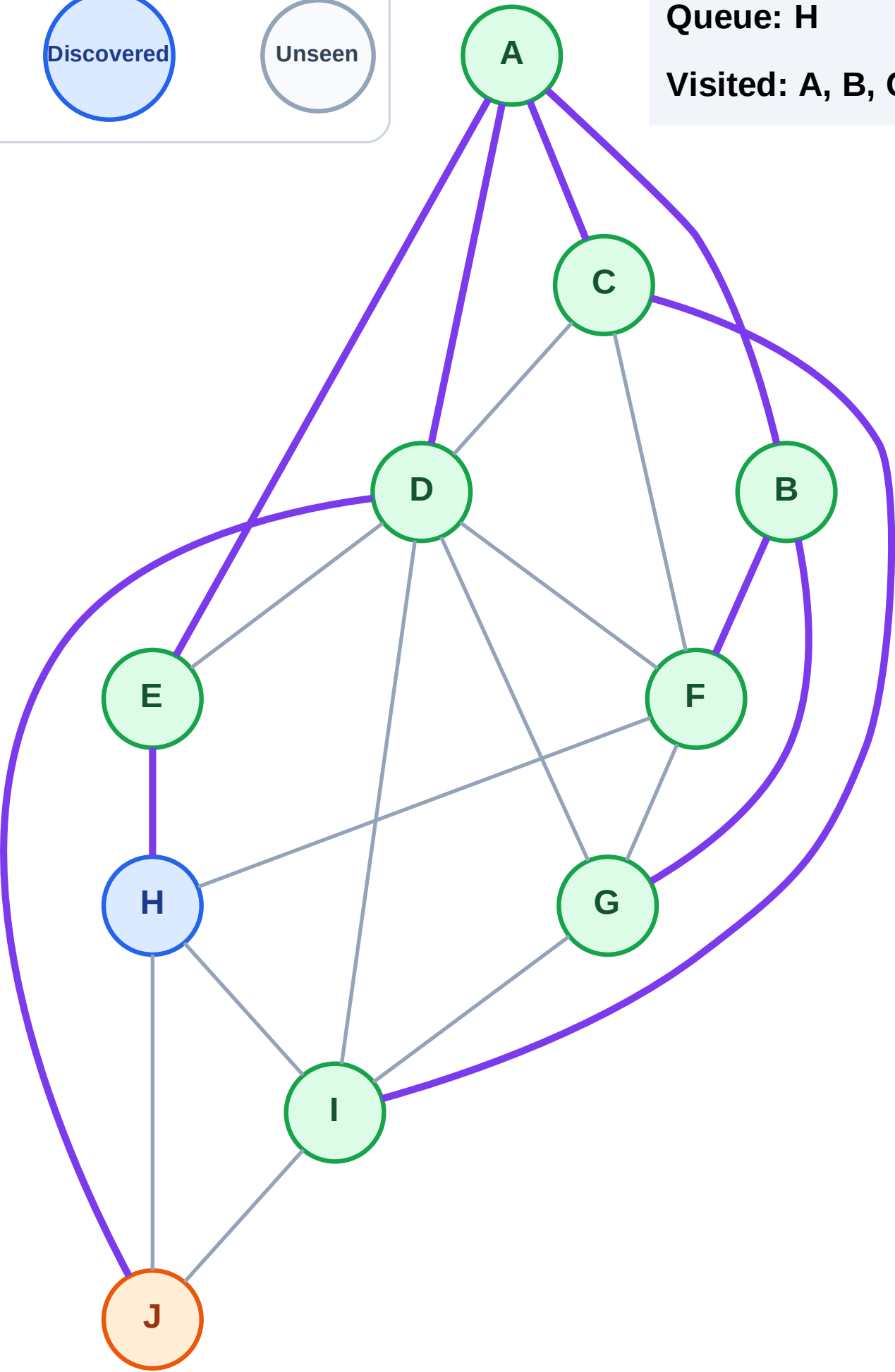
Processing

Discovered

Unseen

Queue: H

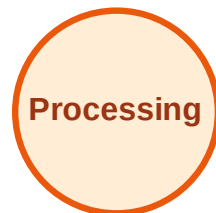
Visited: A, B, C, D, E, F, G, I



BFS Traversal

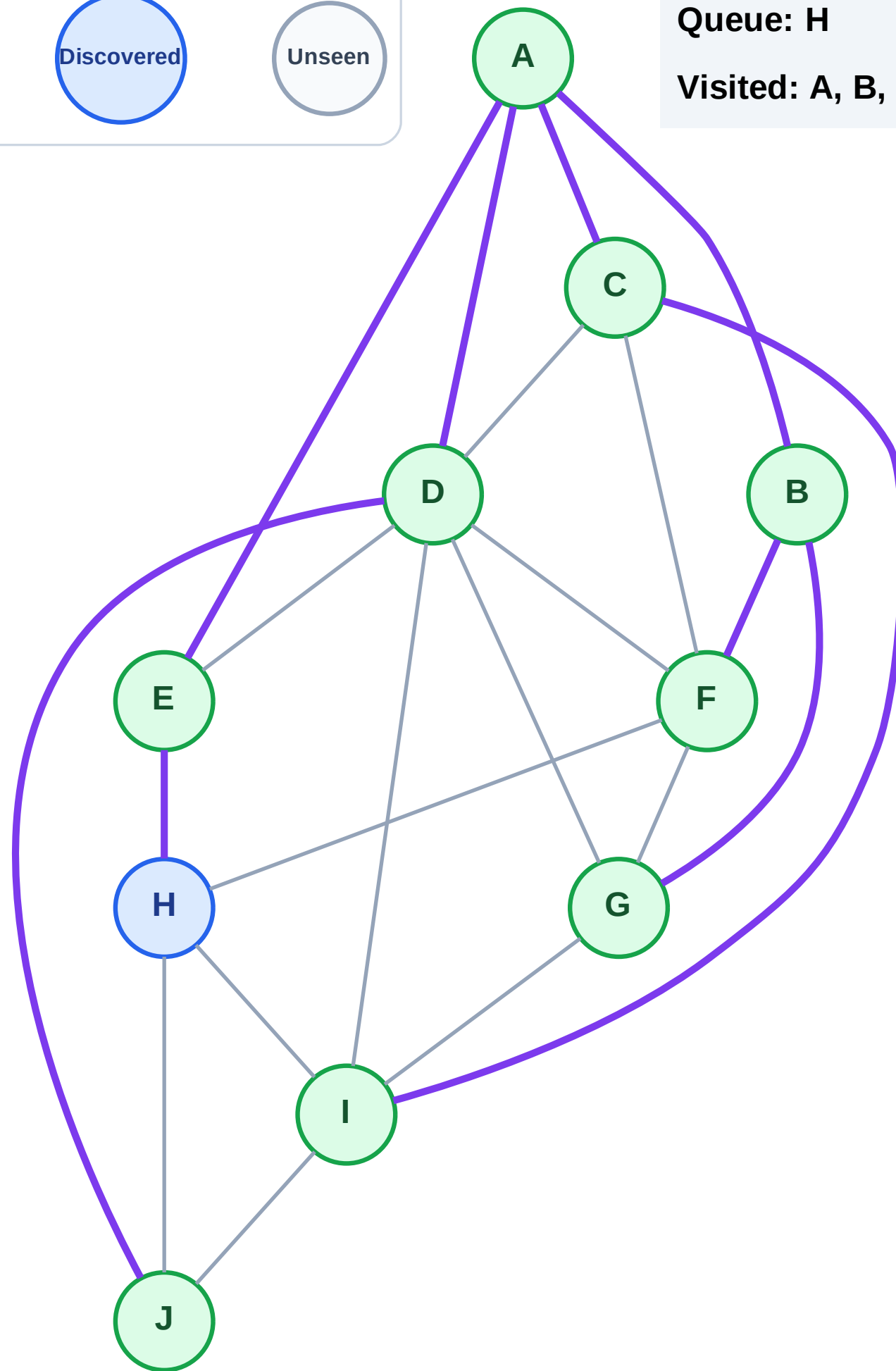
Step 19: No new neighbors from J

Legend



Queue: H

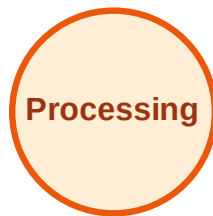
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

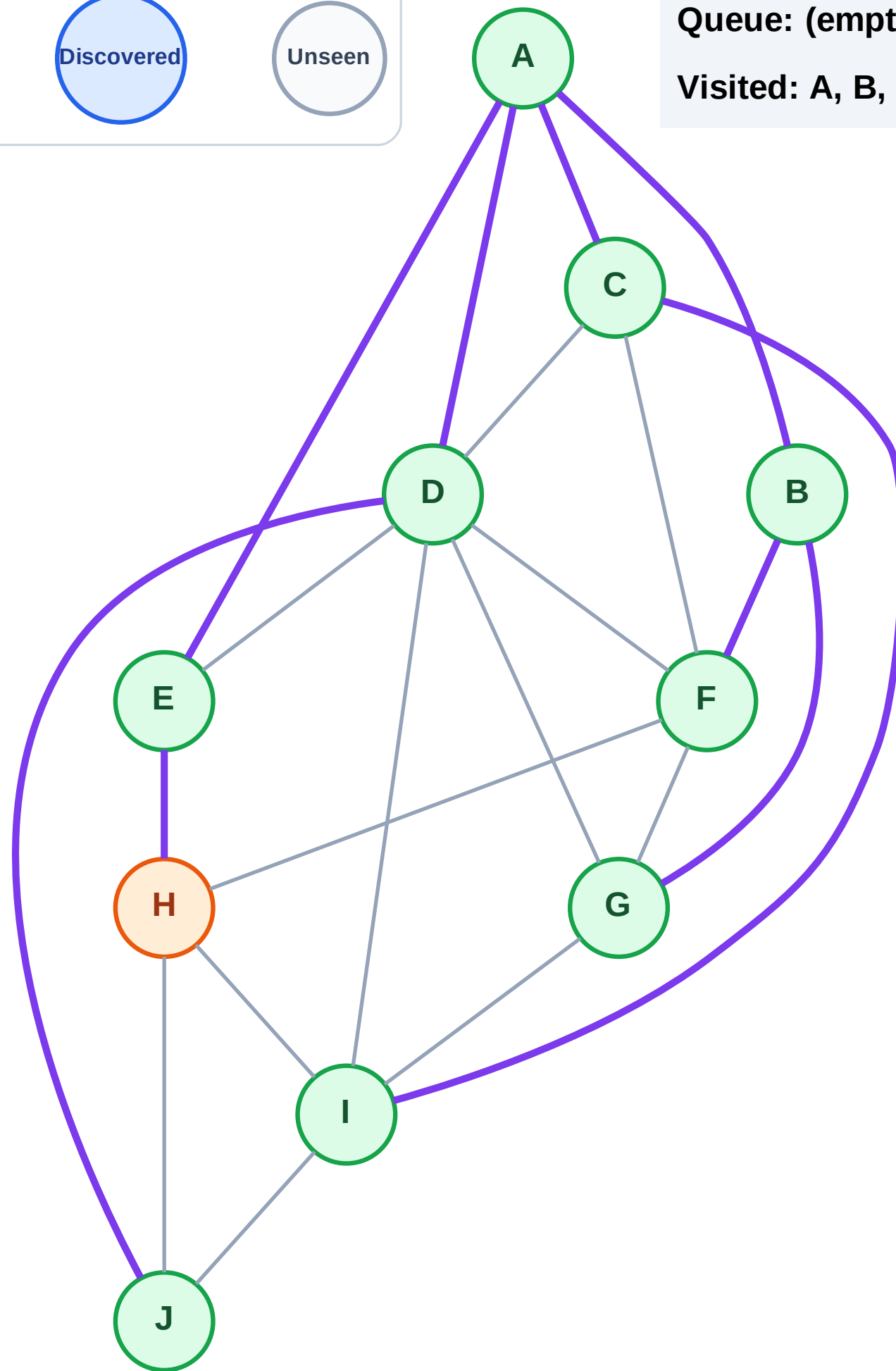
Step 20: Process node H

Legend



Queue: (empty)

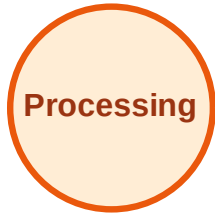
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

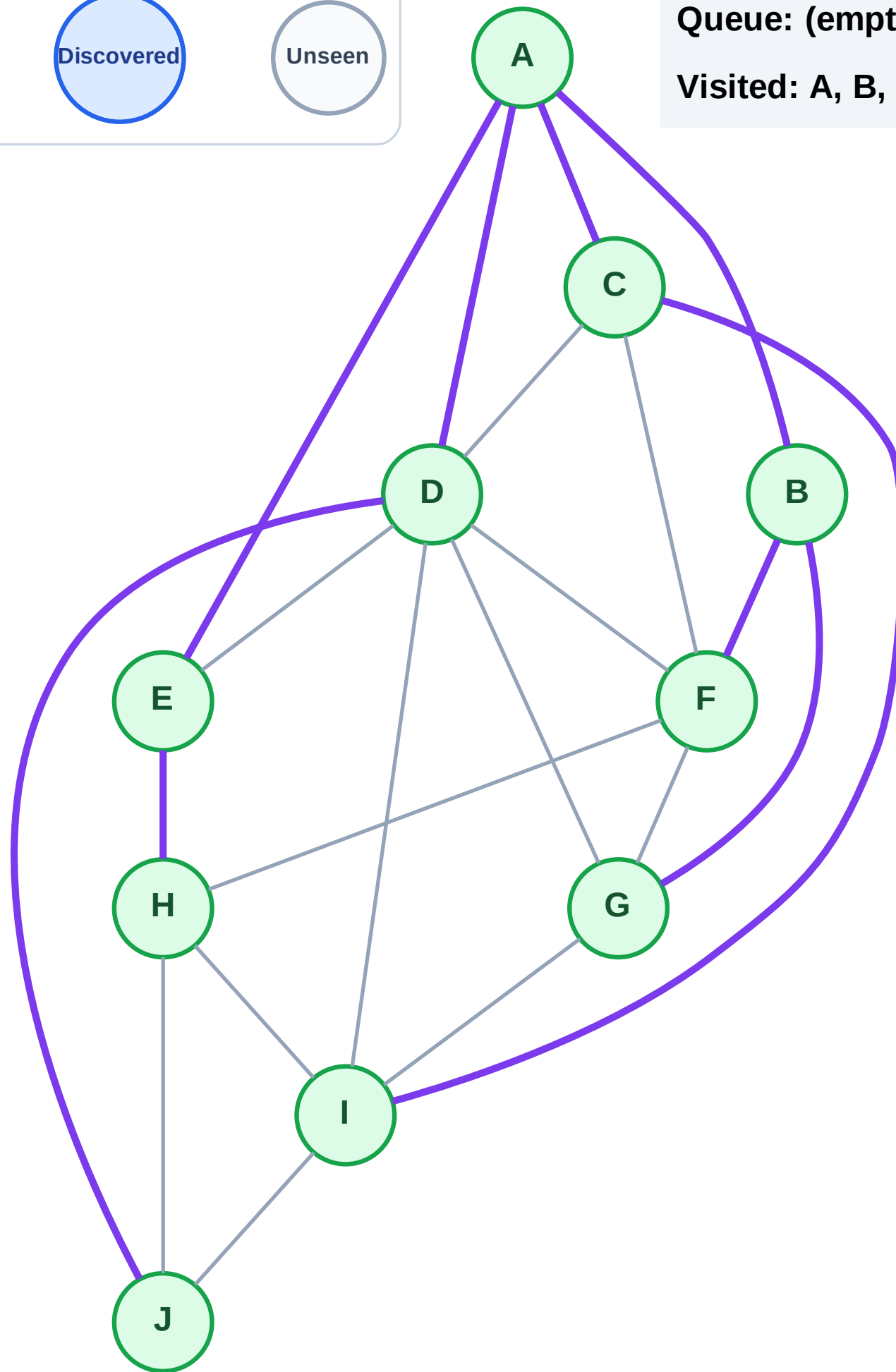
Step 21: No new neighbors from H

Legend



Queue: (empty)

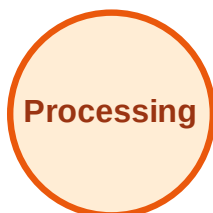
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

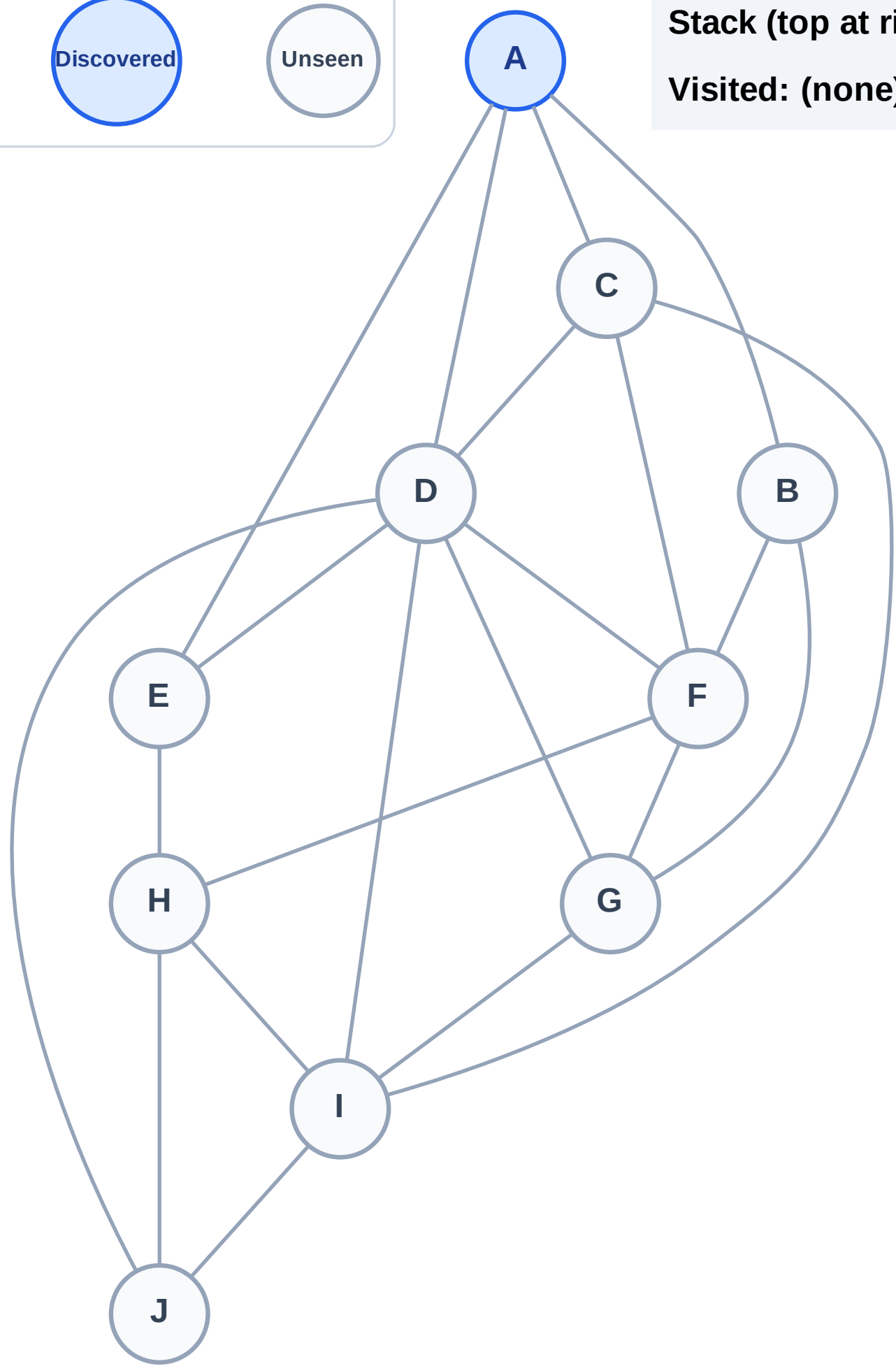
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

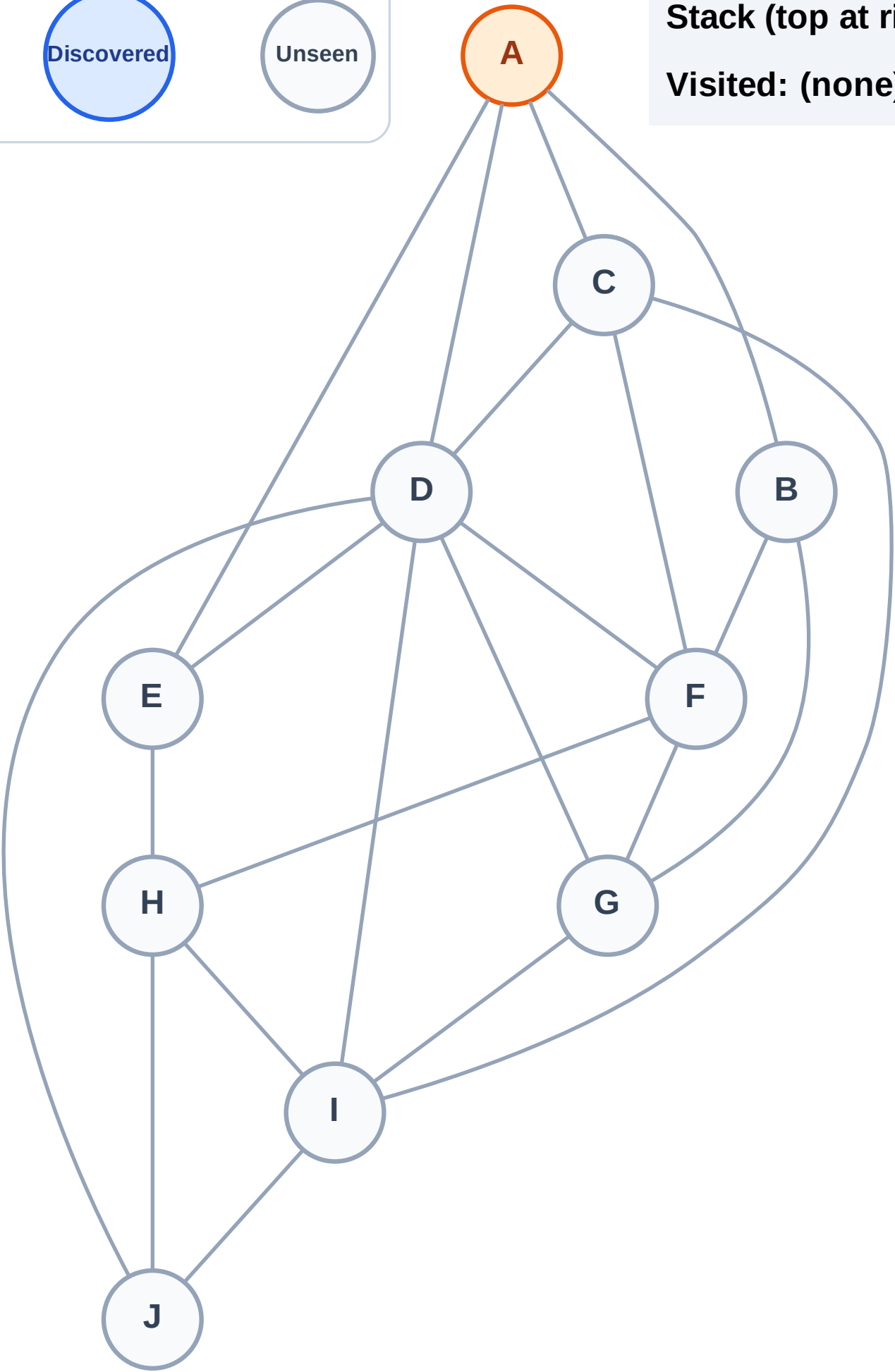
Complete

Processing

Discovered

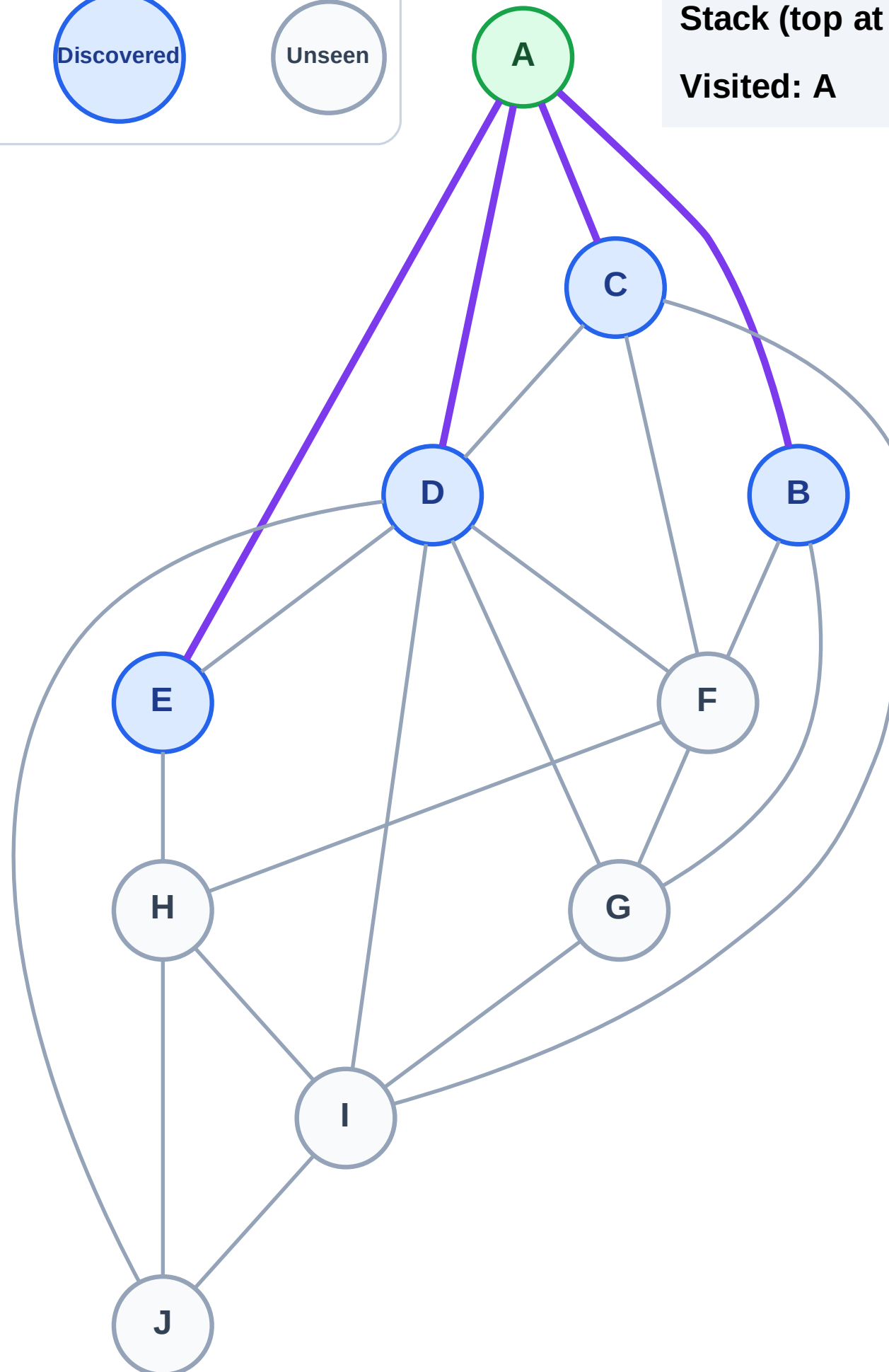
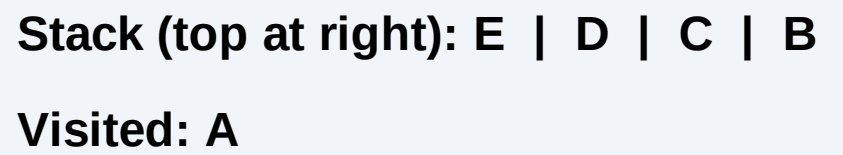
Unseen

Stack (top at right): (empty)
Visited: (none)



Step 3: Discover E, D, C, B from A

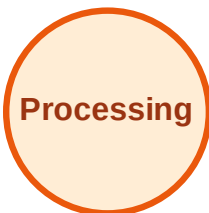
Step 3: Discover E, D, C, B from A



DFS Traversal

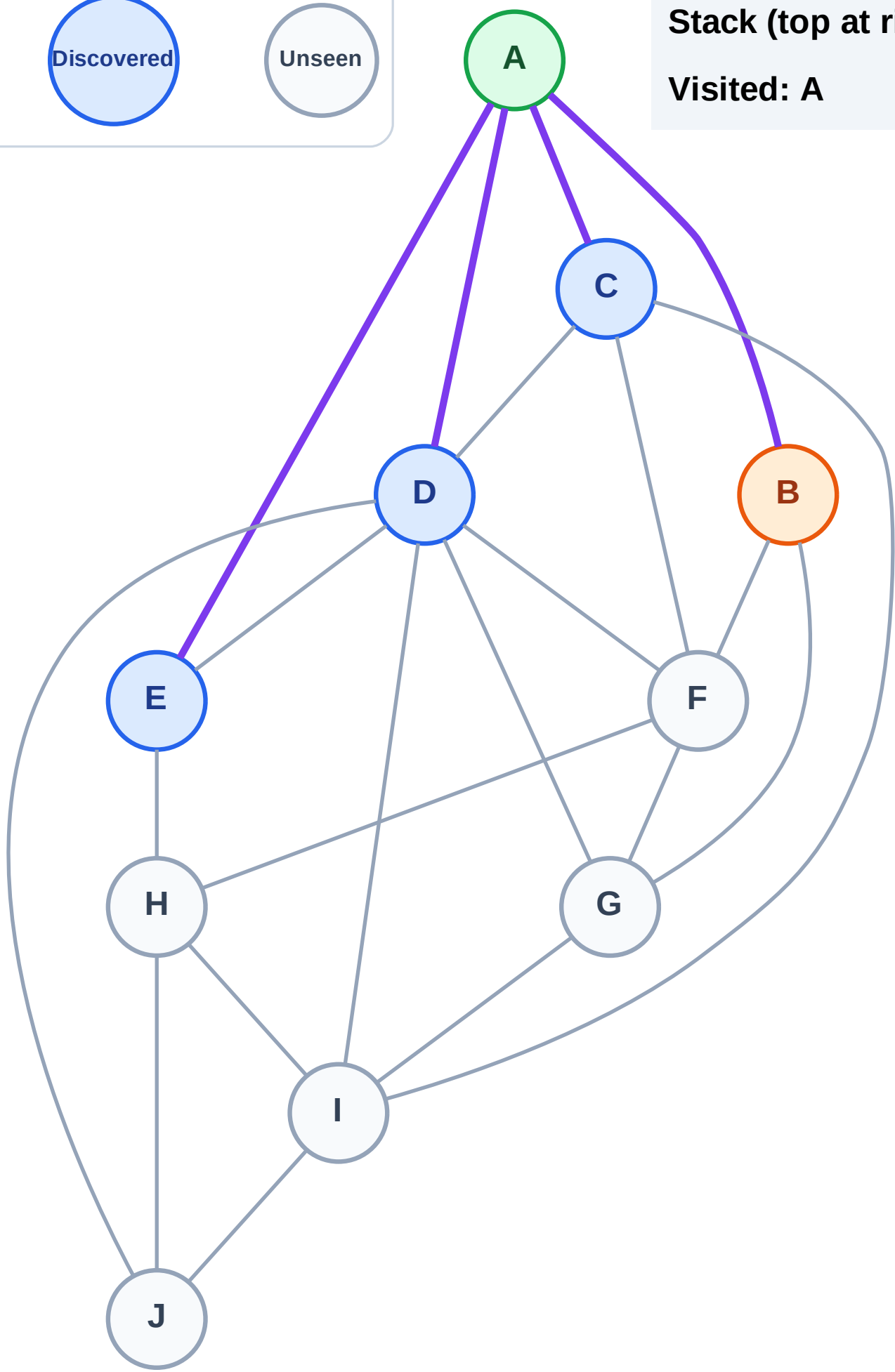
Step 4: Process node B

Legend



Stack (top at right): E | D | C

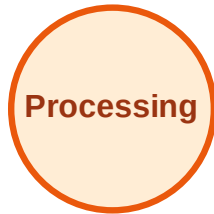
Visited: A



DFS Traversal

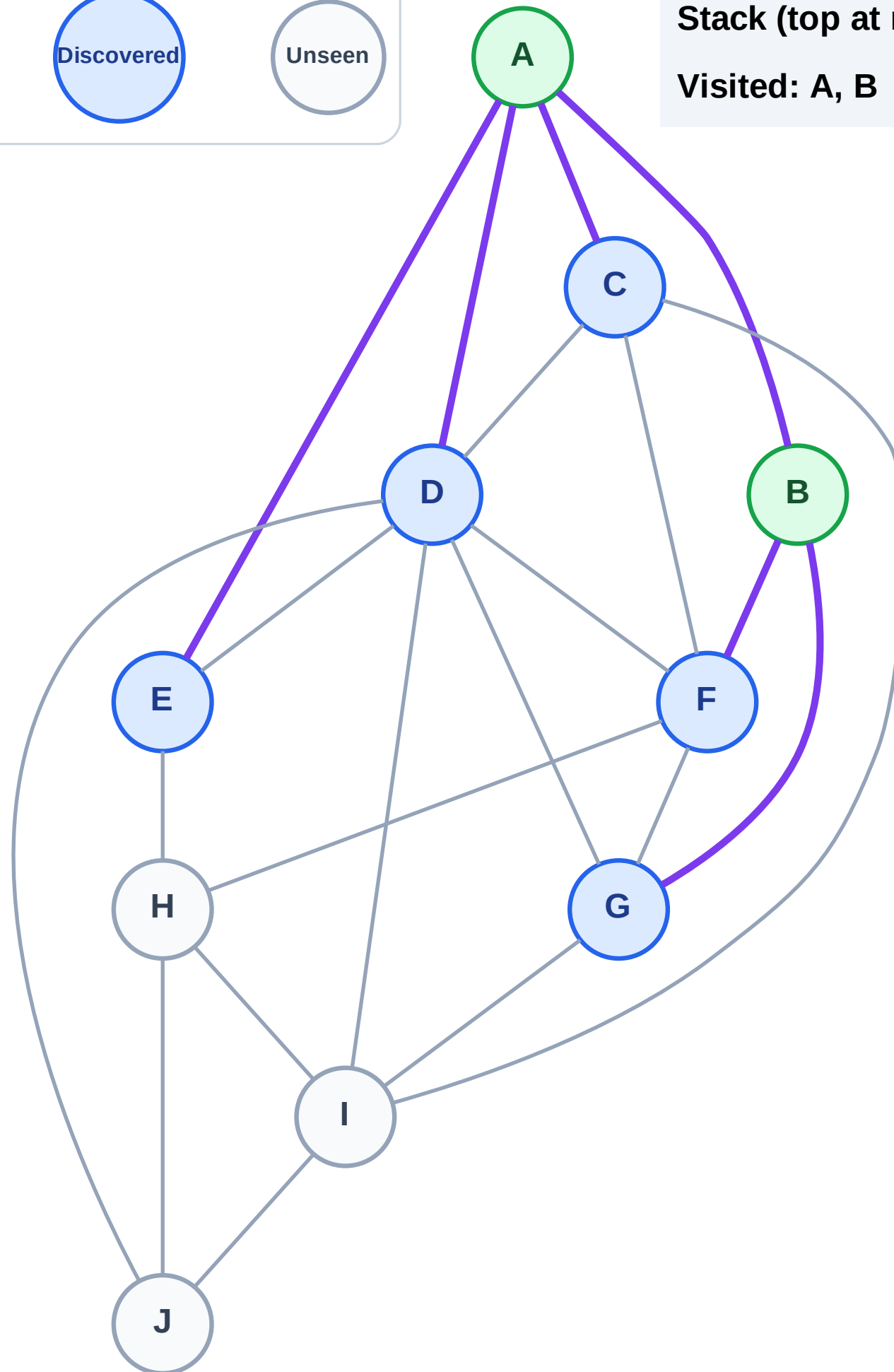
Step 5: Discover G, F from B

Legend



Stack (top at right): E | D | C | G | F

Visited: A, B



DFS Traversal

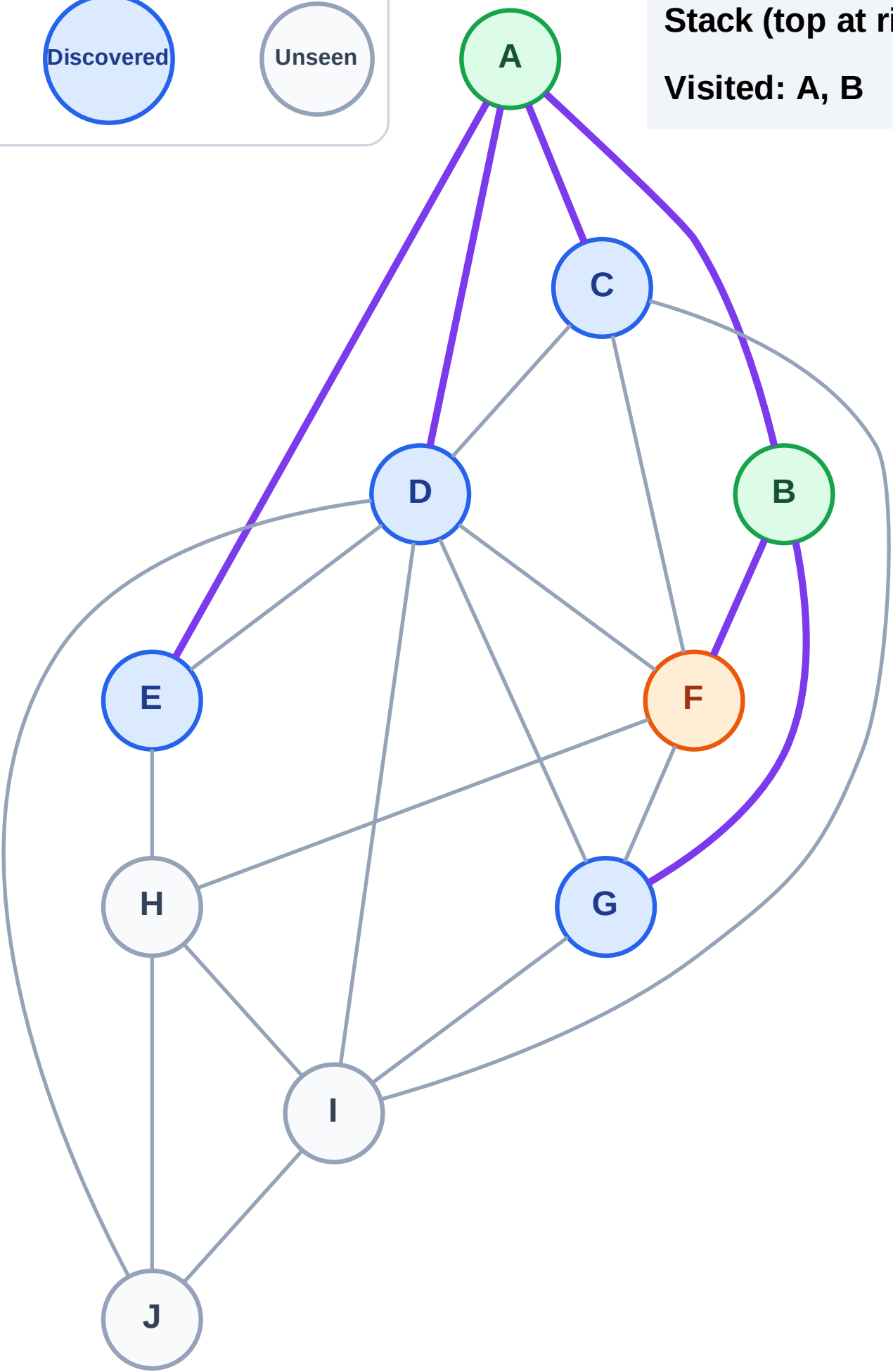
Step 6: Process node F

Legend



Stack (top at right): E | D | C | G

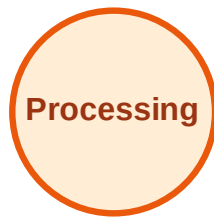
Visited: A, B



DFS Traversal

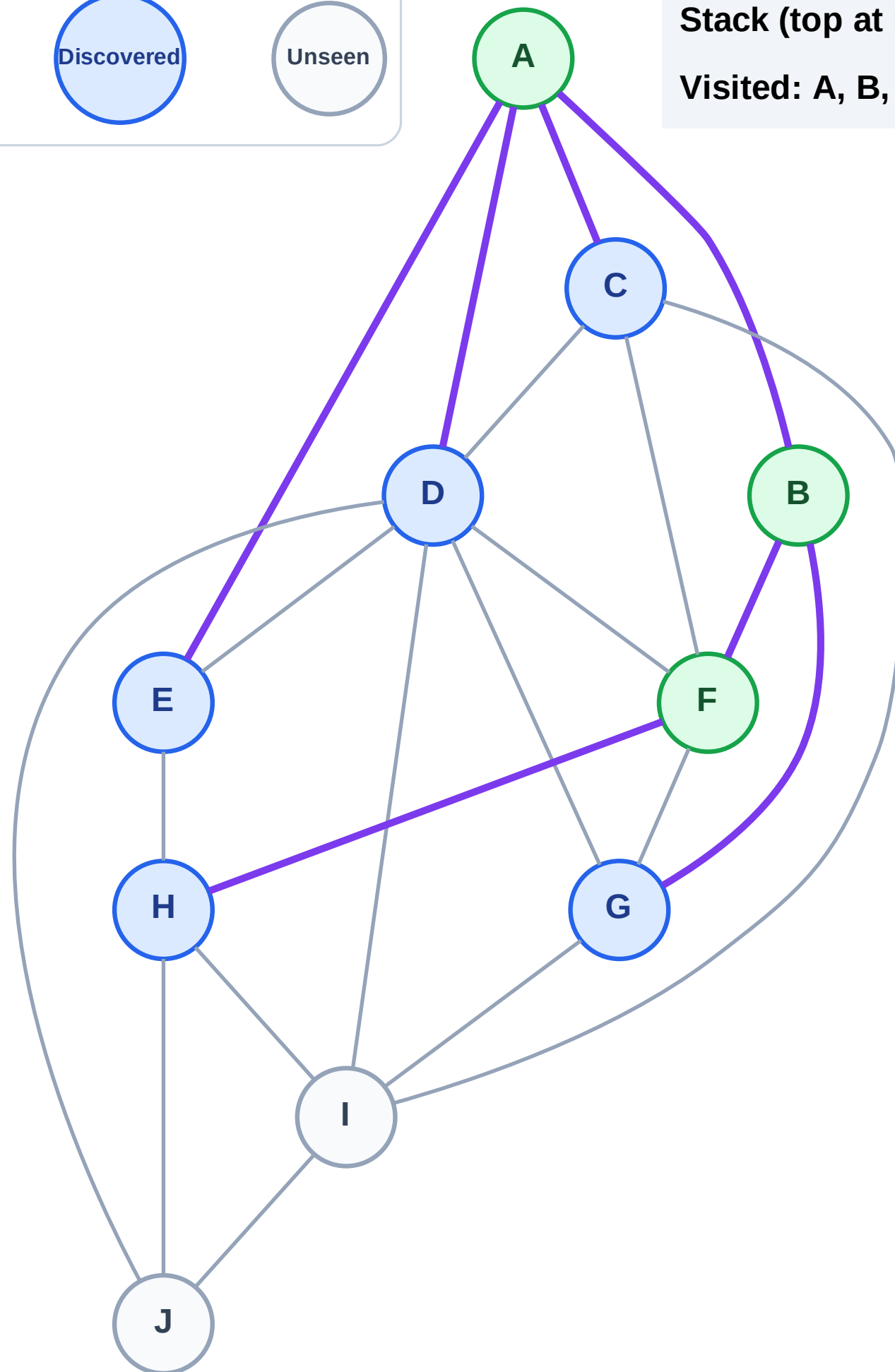
Step 7: Discover H from F

Legend



Stack (top at right): E | D | C | G | H

Visited: A, B, F



DFS Traversal

Step 8: Process node H

Legend

Complete

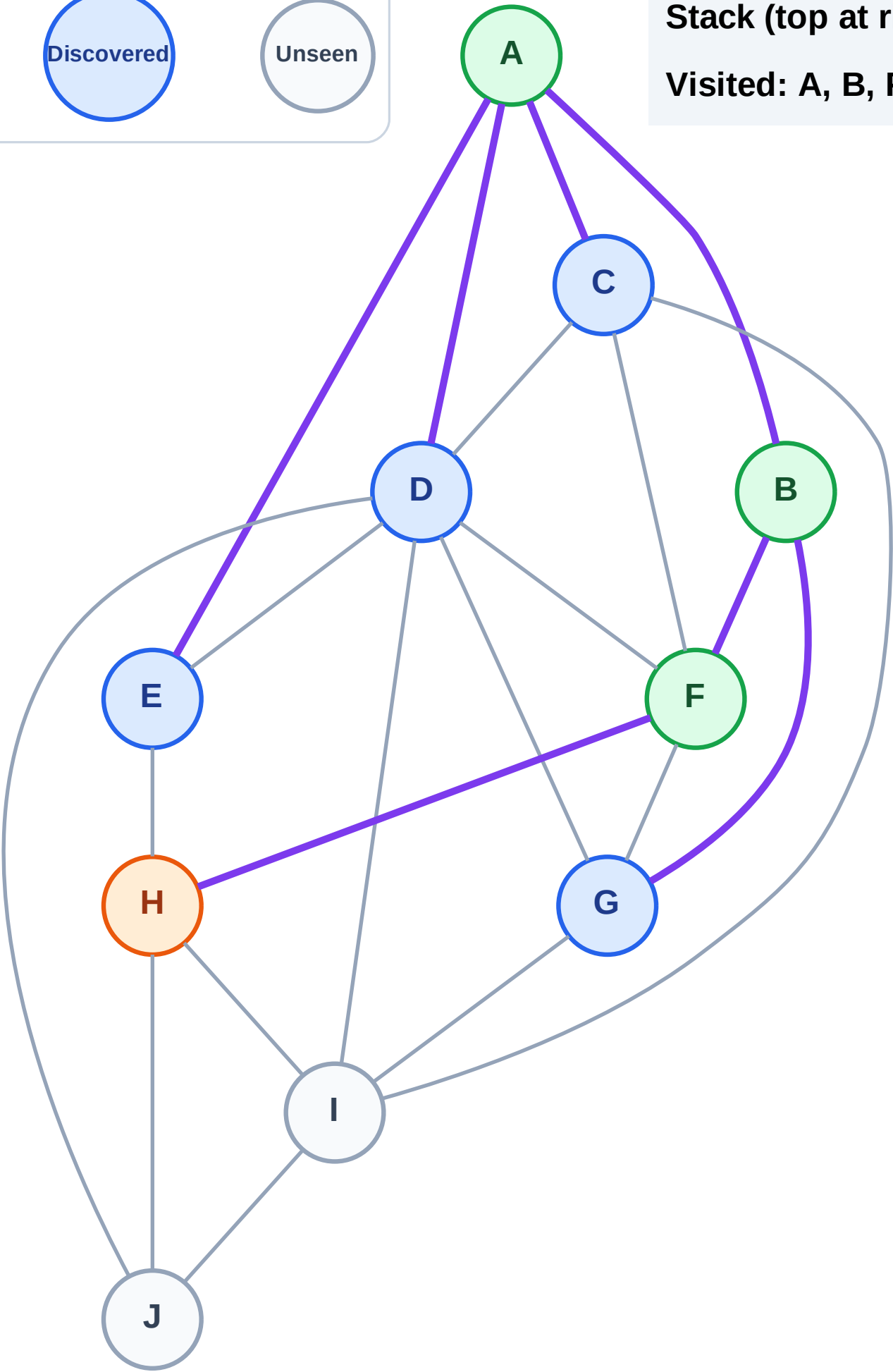
Processing

Discovered

Unseen

Stack (top at right): E | D | C | G

Visited: A, B, F



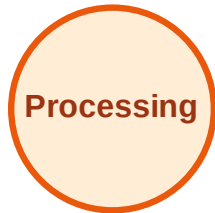
DFS Traversal

Step 9: Discover J, I from H

Legend



Complete



Processing



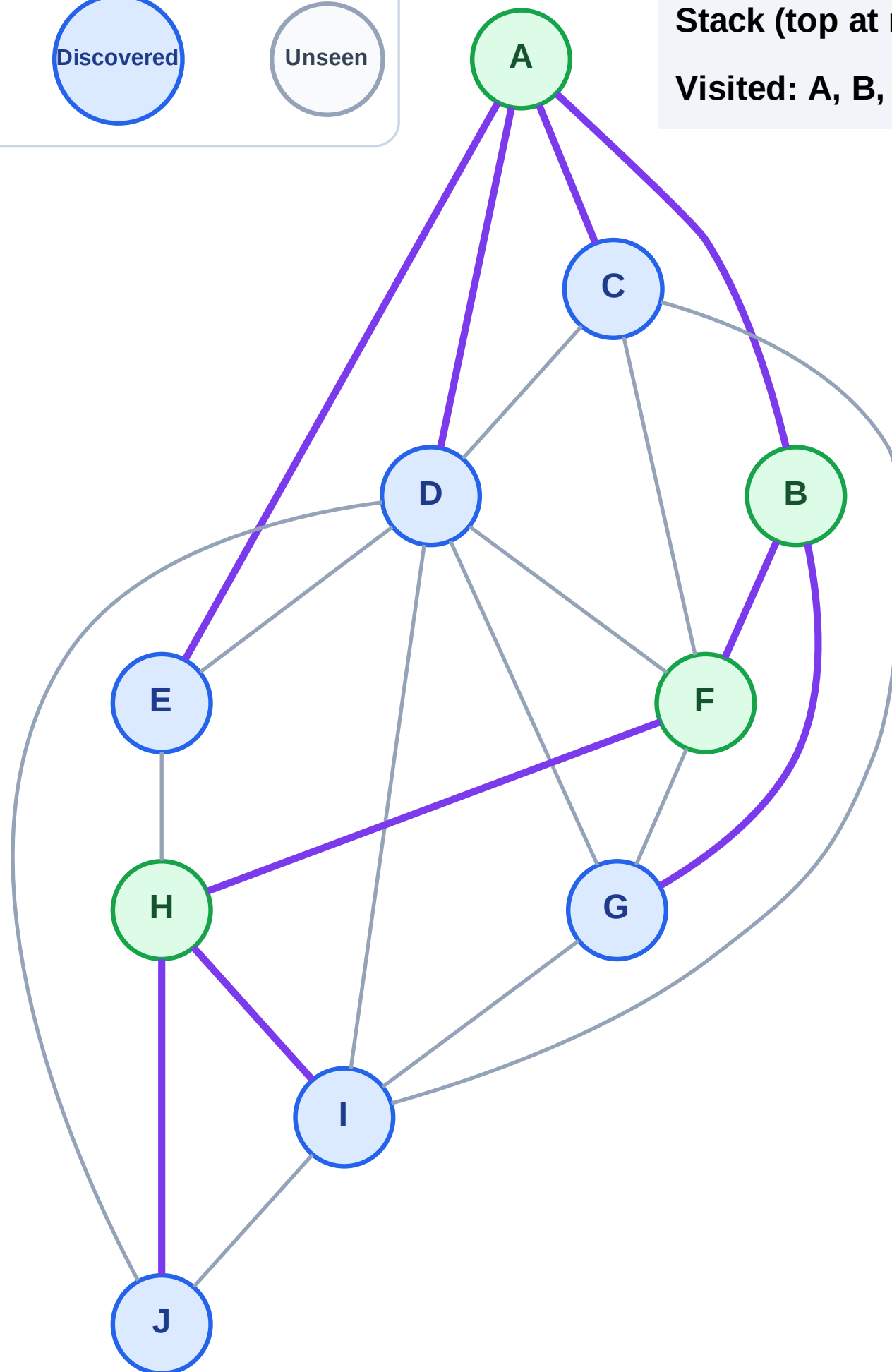
Discovered



Unseen

Stack (top at right): E | D | C | G | J | I

Visited: A, B, F, H



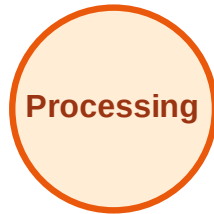
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



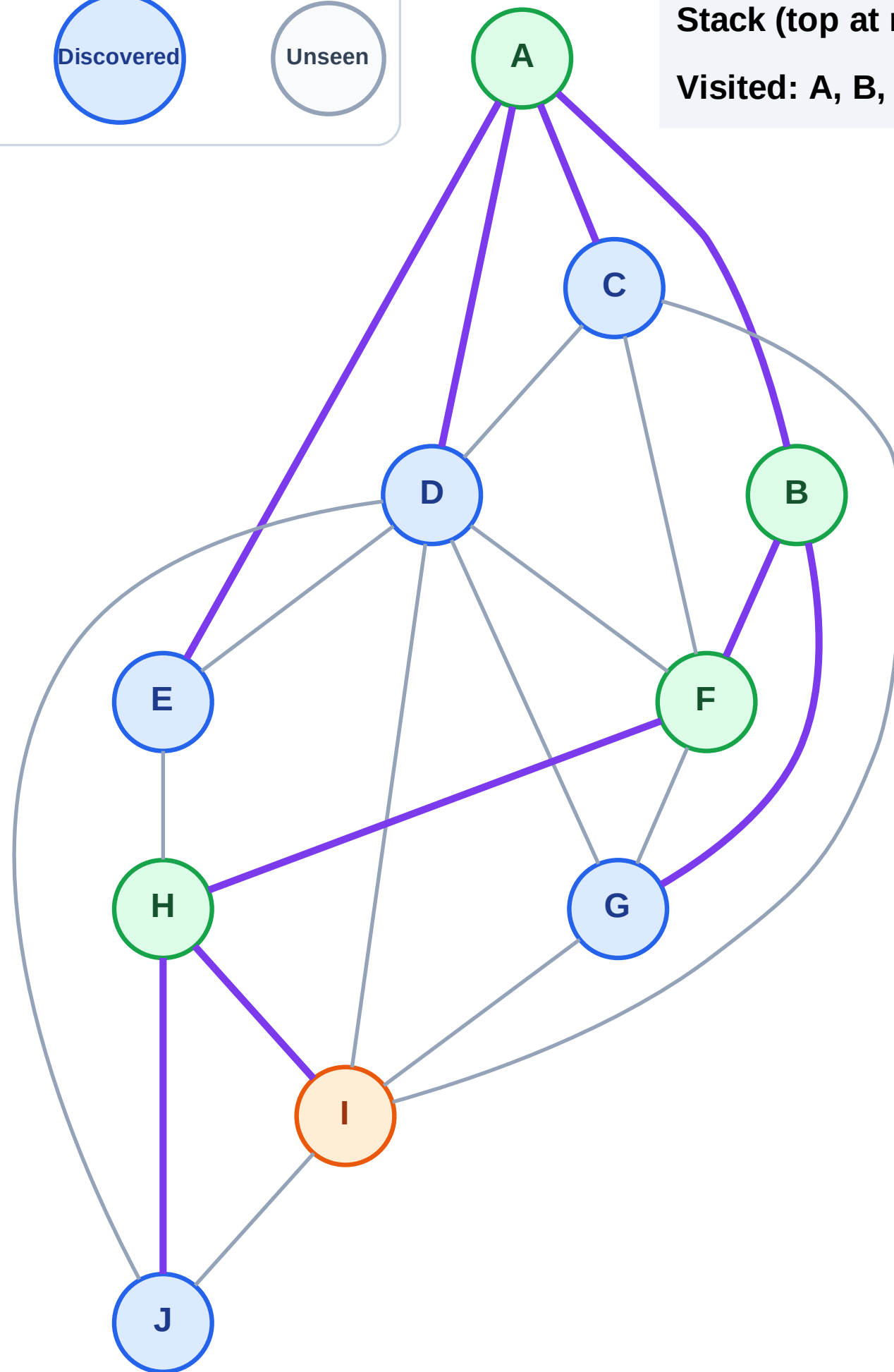
Discovered



Unseen

Stack (top at right): E | D | C | G | J

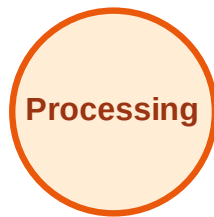
Visited: A, B, F, H



DFS Traversal

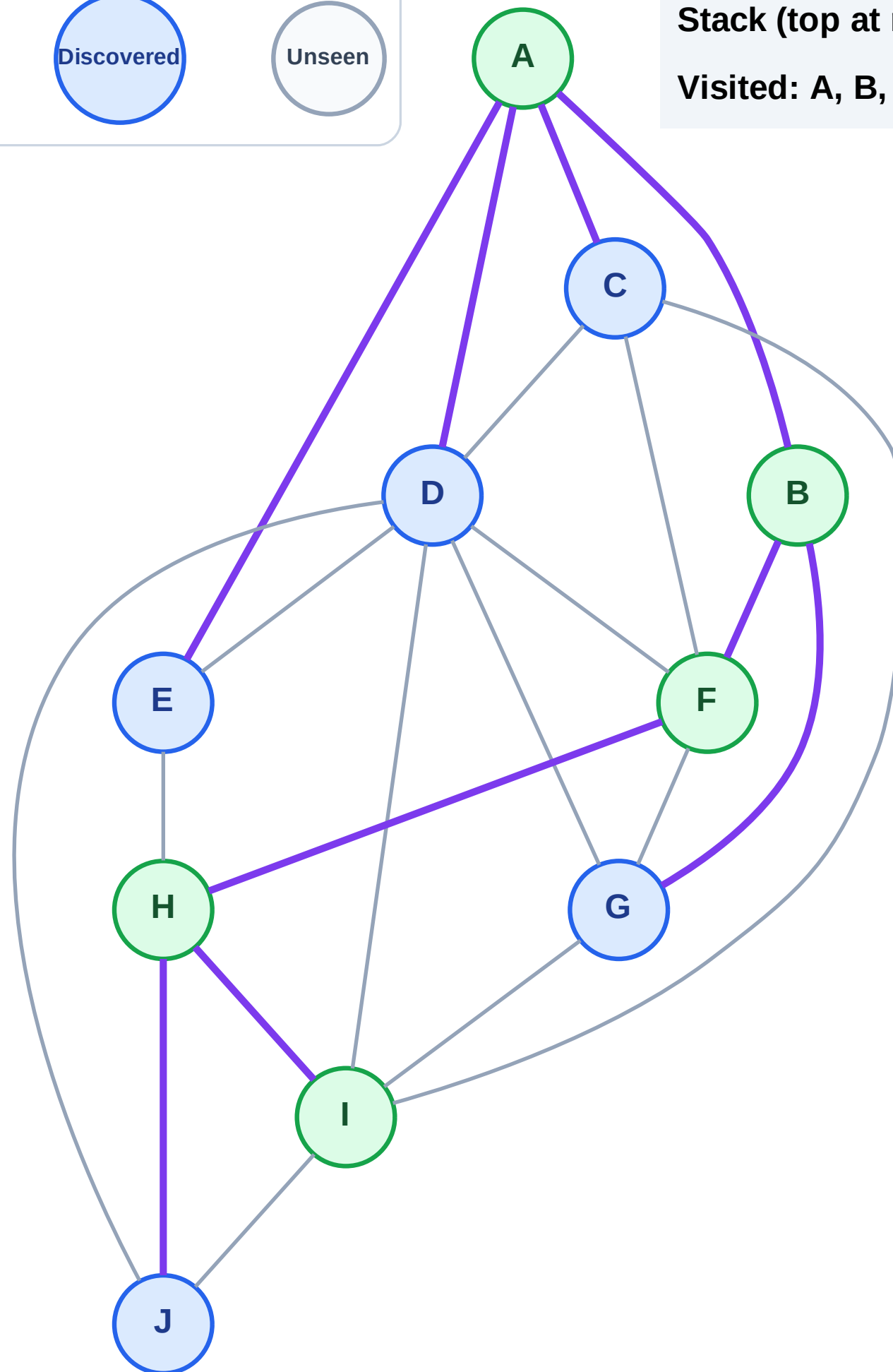
Step 11: No new neighbors from I

Legend



Stack (top at right): E | D | C | G | J

Visited: A, B, F, H, I



DFS Traversal

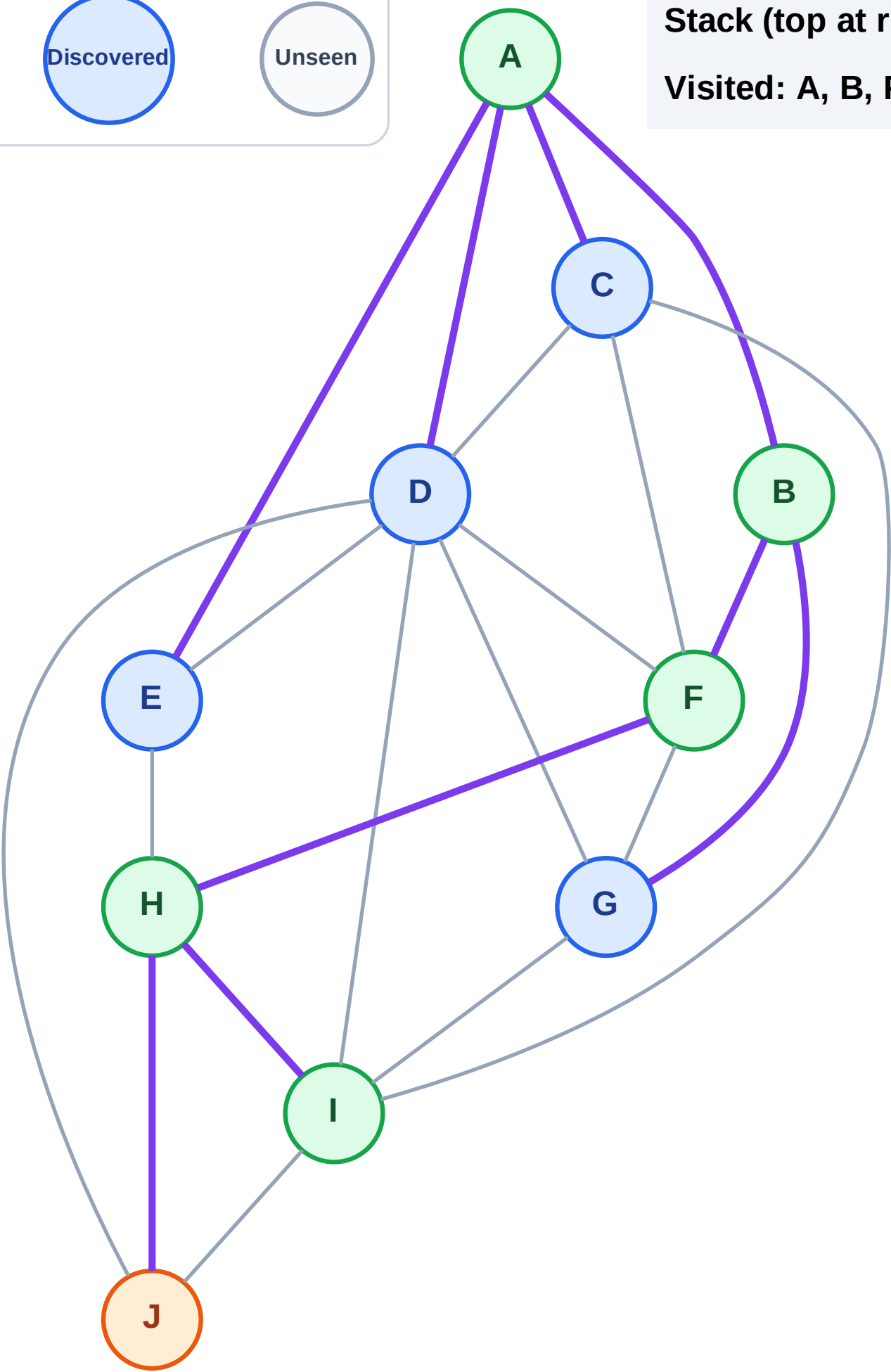
Step 12: Process node J

Legend



Stack (top at right): E | D | C | G

Visited: A, B, F, H, I



DFS Traversal

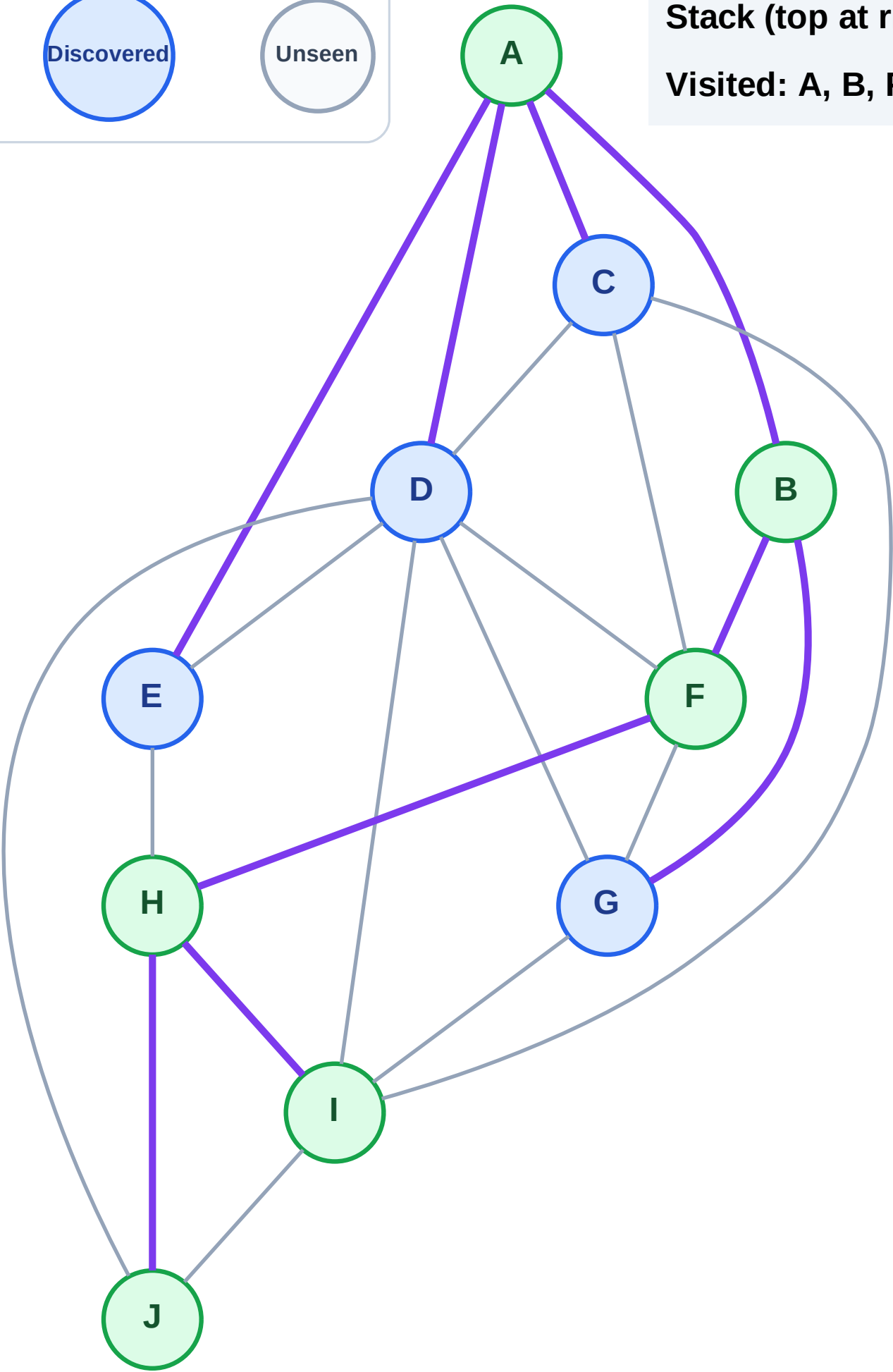
Step 13: No new neighbors from J

Legend



Stack (top at right): E | D | C | G

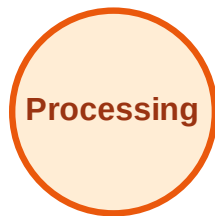
Visited: A, B, F, H, I, J



DFS Traversal

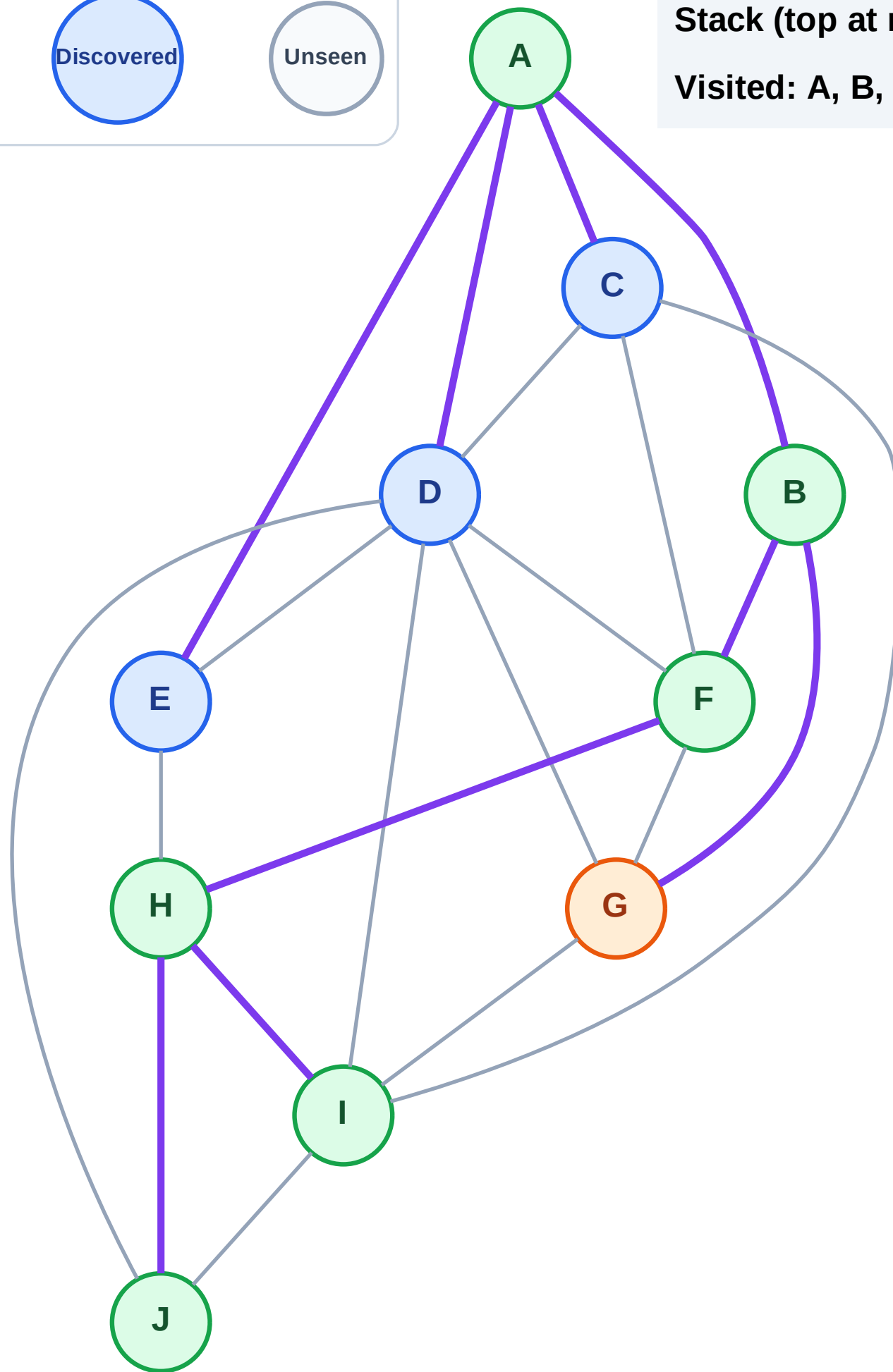
Step 14: Process node G

Legend



Stack (top at right): E | D | C

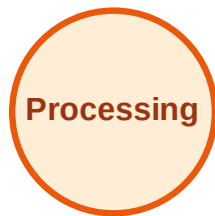
Visited: A, B, F, H, I, J



DFS Traversal

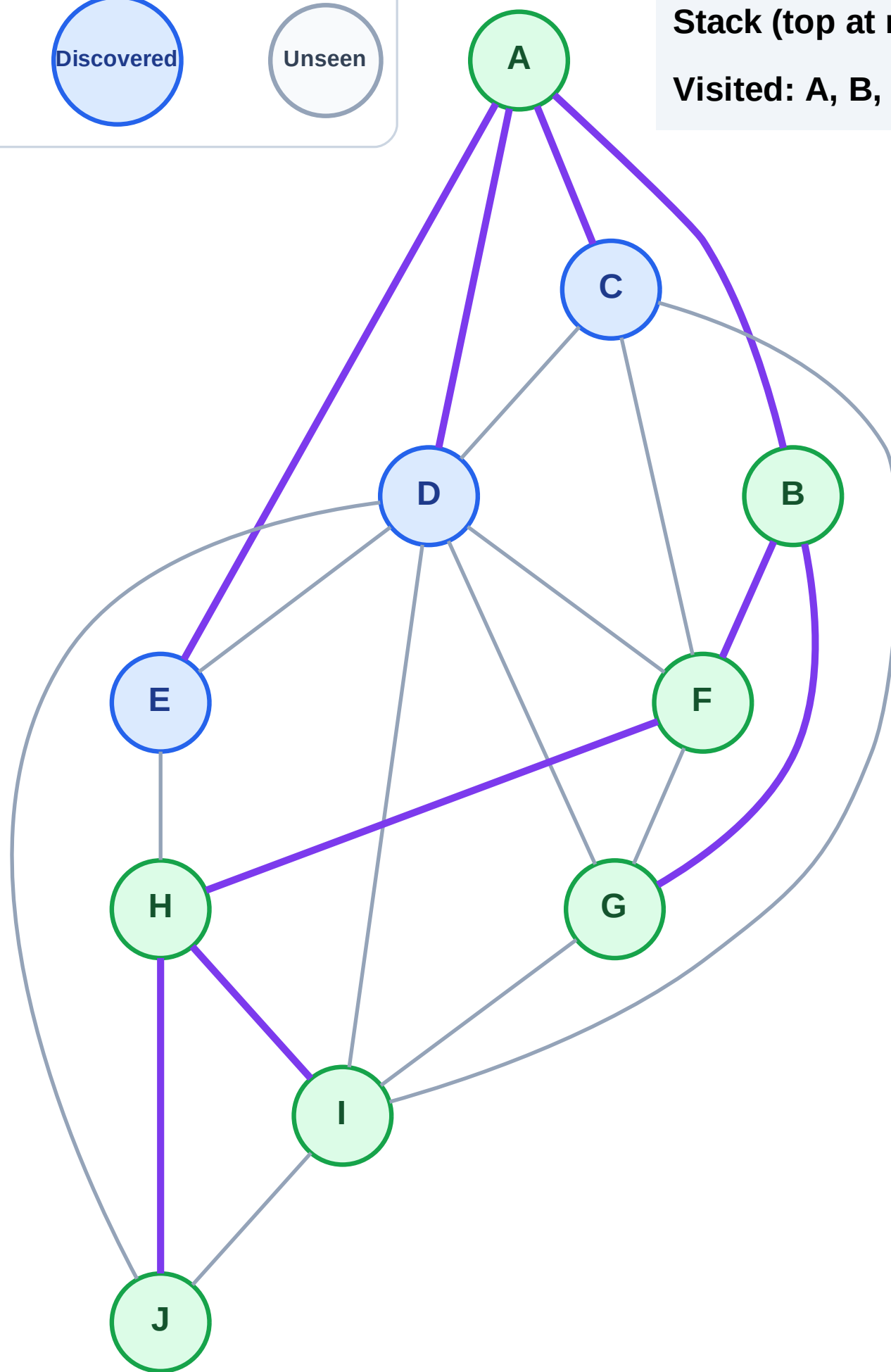
Step 15: No new neighbors from G

Legend



Stack (top at right): E | D | C

Visited: A, B, F, G, H, I, J



DFS Traversal

Step 16: Process node C

Legend

Complete

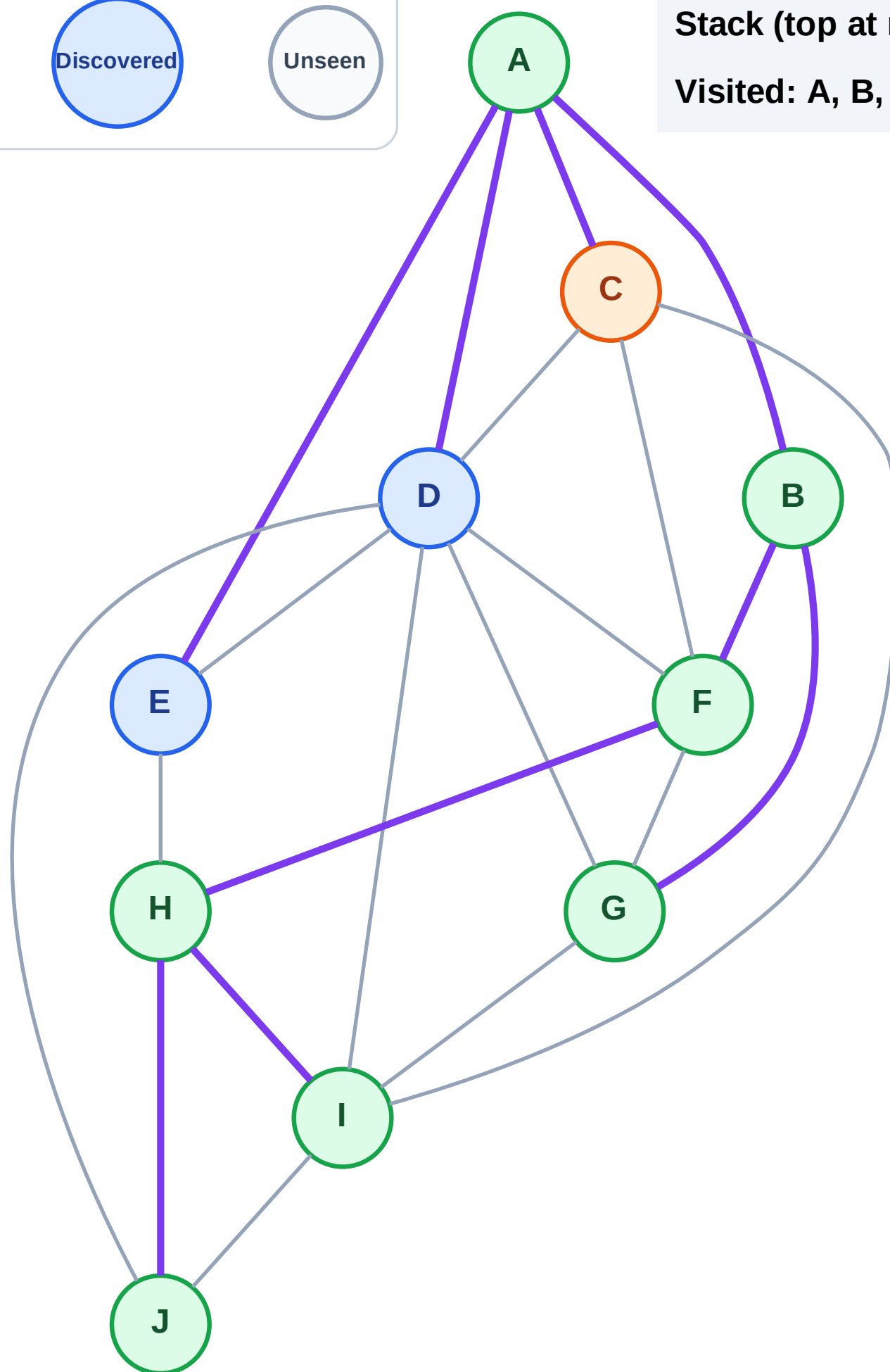
Processing

Discovered

Unseen

Stack (top at right): E | D

Visited: A, B, F, G, H, I, J

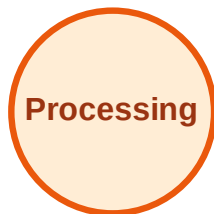


Step 17: No new neighbors from C

Step 17: No new neighbors from C

Complete

Complete



Processing

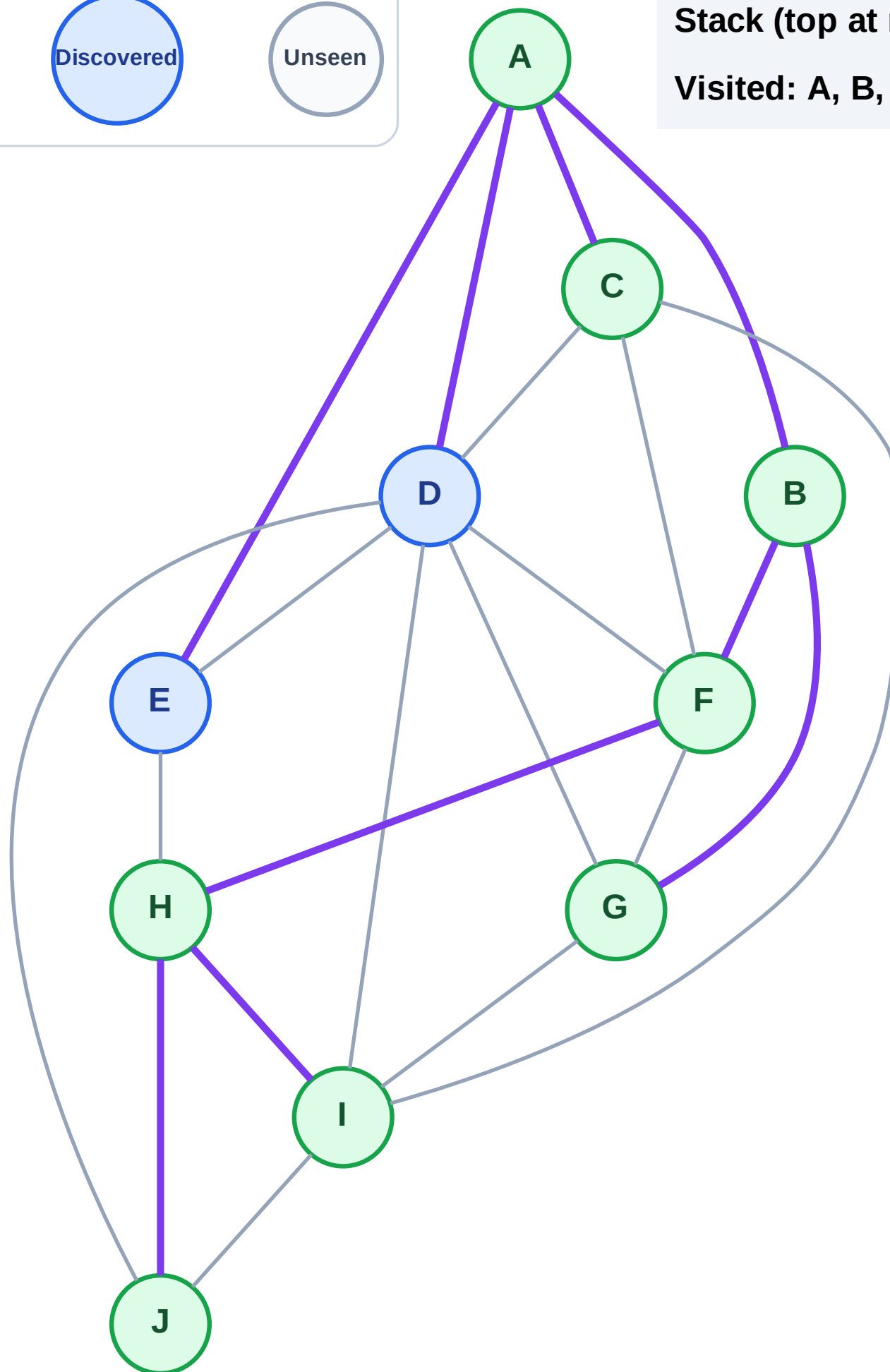


Discovered



Unseen

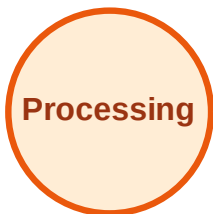
Visited: A, B, C, F, G, H, I, J



DFS Traversal

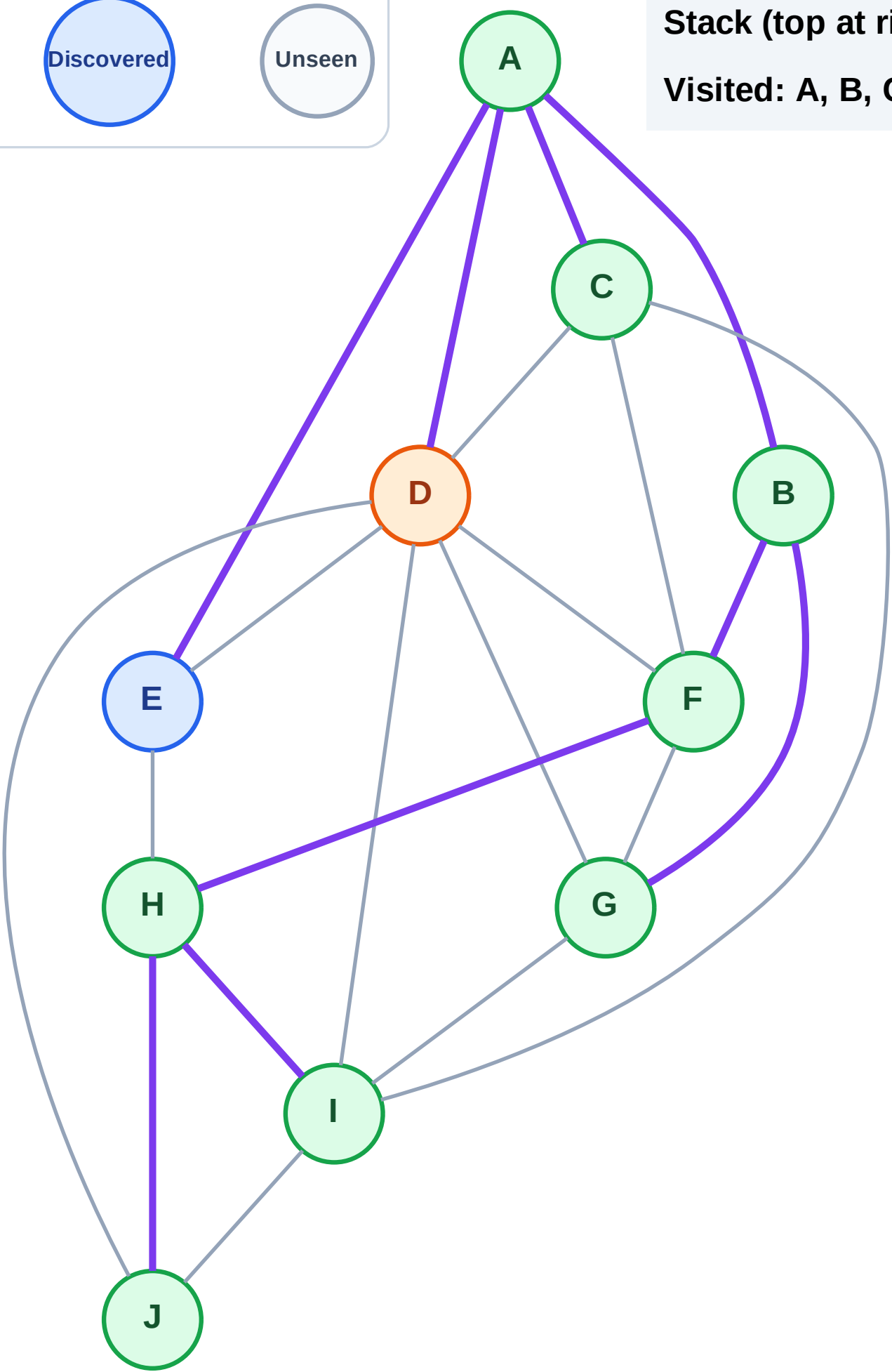
Step 18: Process node D

Legend



Stack (top at right): E

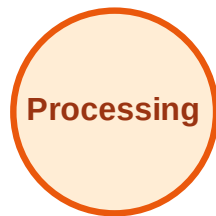
Visited: A, B, C, F, G, H, I, J



DFS Traversal

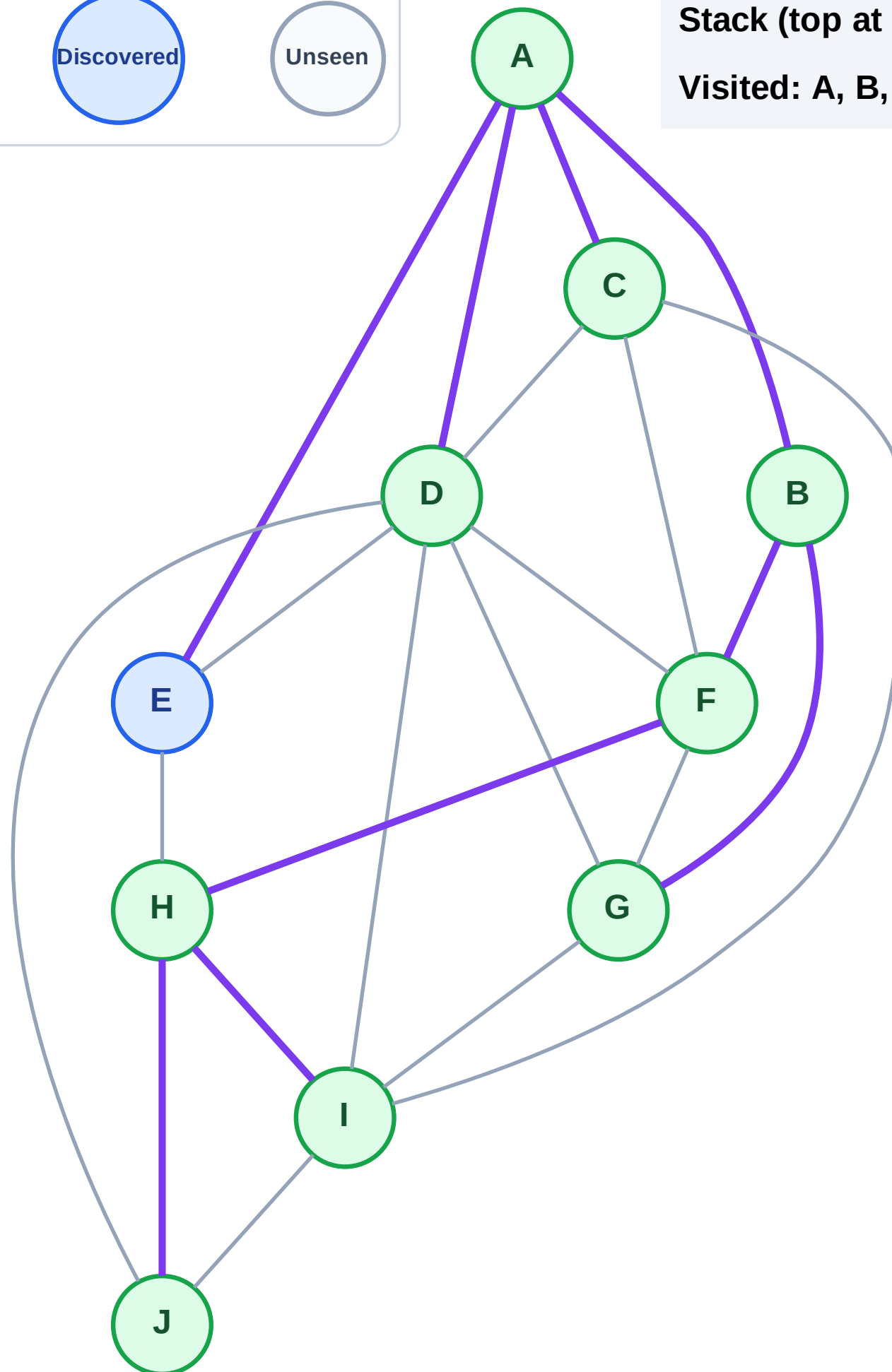
Step 19: No new neighbors from D

Legend



Stack (top at right): E

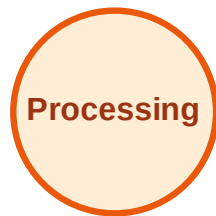
Visited: A, B, C, D, F, G, H, I, J



DFS Traversal

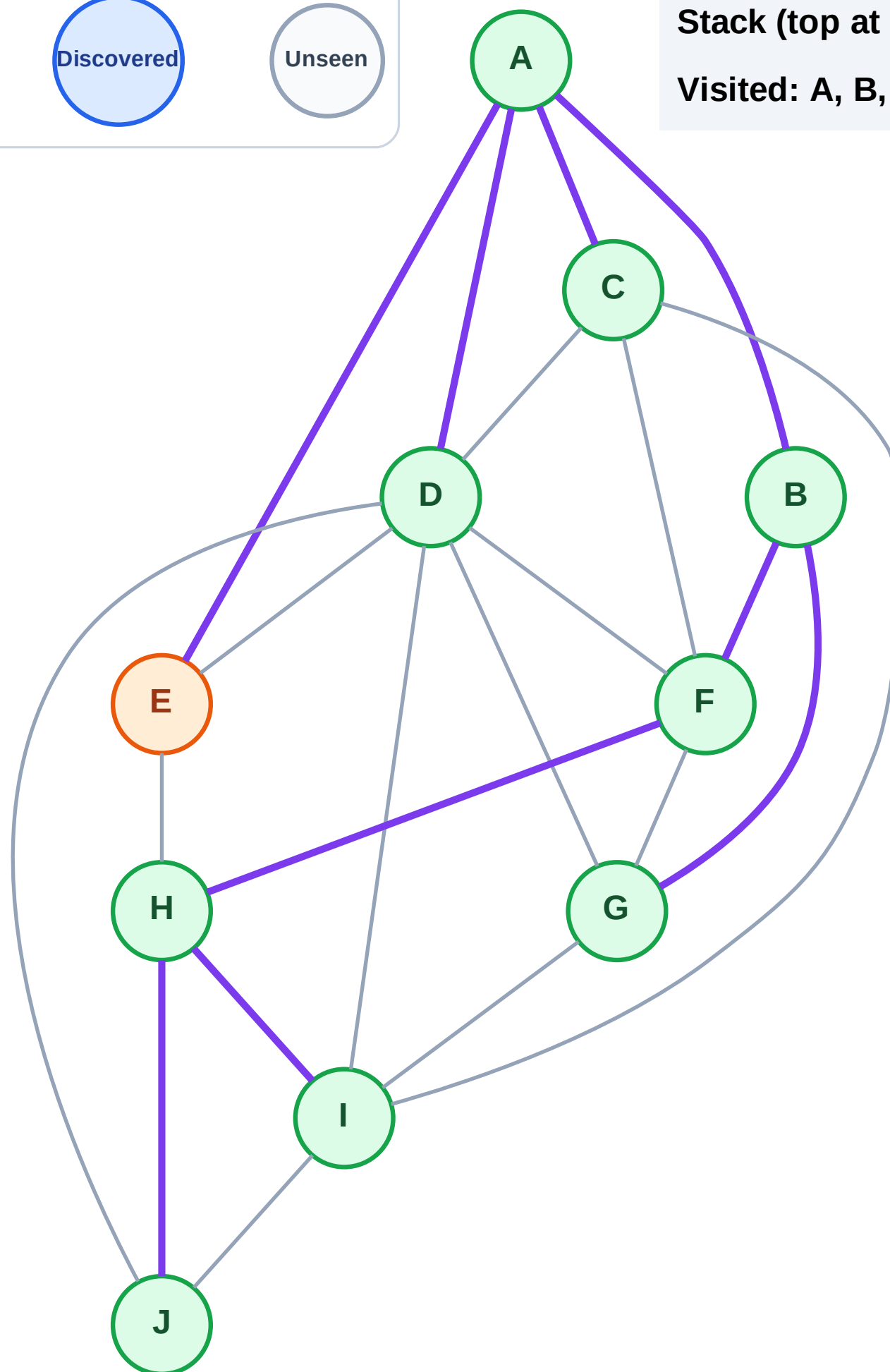
Step 20: Process node E

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from E

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

